Synthetic Bengali Handwriting: Learned Page Layout and Writer-Conditioned Word Generation

Final technical report

scribe_layout / scribe_layout_v2 / scribe_wordgen projects

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Abstract

BengaliScribe synthesises handwritten Bengali pages by generating word images with a font-prior GAN (baseline *B0*) and pasting them onto a page with hand-picked layout constants. Its pages look *typeset in a handwriting-like font*: the words re-ink a NotoSans skeleton with a stroke about 55% wider than the target writer's, and the lines are evenly spaced and straight. This report covers two lines of work. (1) **Page layout.** We measured how real writers lay out words on 963 full-page scans and trained a small page-autoregressive mixture-density model (v1, then v2). The final default (v2 `default_unfitw_dc`) reaches a plan-level line C2ST AUC of 0.535 ± 0.012 on unseen texts (v1: 0.566 ± 0.021; hand-tuned layout: 0.980; within-page null 0.496). Two spectral side-studies are reported with their corrections. (2) **Word generation.** We built a few-shot latent-diffusion word generator from DiffusionPen and a full evaluation suite. That suite has a locator CER, an independent CRNN recogniser (R_eval), writer-ID style metrics (E_eval/E_shape) and a sub-pixel pen metric. All statistics use writer-bootstrap CIs. The best diffusion arm, **S** (pen knob plus reference tokens in decoupled cross-attention), ties B0 on locator CER and on style MRR (0.579 vs 0.596), with a matched pen after calibration. Under R_eval it is still *less legible* than B0 ($+0.033$ [+0.015, +0.052] codepoint CER on words of ≥ 3 codepoints). Pretraining R_eval on renders from 48 Bengali fonts helps a little with full data and a lot with 10% of the writers. Every number in this report is traced to a result file. Three experiments still running are listed only as next steps.

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1 Executive summary

Key findings.

- **B0's "font look" is architectural, not a training accident.** B0 keeps fine-resolution skips from a NotoSans prior. Its only deformation module can move ink at most 1.4 px vertically on a 64-px canvas. Its losses reward the averaged font skeleton. Generated page strokes are 0.39 mm wide against 0.25 mm real (§2).
- **The learned layout beats hand-tuned constants by a wide margin and v1 by a small one.** The plan-level line C2ST is 0.980 for manual, 0.566 ± 0.021 for v1 and 0.535 ± 0.012 for the v2 default on TEST (0.536 ± 0.027 on VAL). End to end, through BengaliScribe and the real detector, v2 scores a line C2ST of 0.719 (0.619 with writer context), against 0.905 for manual and a floor of 0.534 (§5).
- **Several layout gains were measurement artefacts and were corrected.** Corrected items are a biased end-to-end C2ST (floor 0.34 instead of 0.5), per-word wobble overshoot, selection on TEST, and a Fourier "17× better" claim. In the fair comparison, Fourier and v1 are equal, 0.668 vs 0.702 (§5.5).
- **Diffusion fixes the texture but had to earn legibility.** The first DiffusionPen probe wrote round, flowing, handwriting-like strokes, but its locator CER was 0.65–0.97. Content tokens in cross-attention, full data, white-paper normalisation and a fixed classifier-free-guidance (CFG) null brought B′ to 0.188 at g3. That fix was a bug fix: content had sat in both CFG branches. A guidance of t5c5 then reached 0.148, against 0.130 for B0 (§6).
- **Arm S is the best diffusion arm.** It ties B0 on locator CER ($+0.007$ [-0.023, +0.040]) and on style M2 MRR (-0.017 [-0.116, +0.088]). It beats P on style by $+0.136$ [+0.085, +0.187] (17/20 writers). Its pen is 1.13 raw and 1.06 calibrated (§6.9).
- **R_eval changes the CER picture.** The locator was too weak to separate arms. With R_eval, S−B0 is $+0.059$ [+0.036, +0.083] on all words and $+0.033$ [+0.015, +0.052] on words of $\geq$3 codepoints. Part of the all-word gap is a short-word `img_to_ink` artefact; the $\geq$3-codepoint gap remains significant (§7.3).
- **Recogniser CER measures legibility, not fidelity.** B0 scores *better than the real crops* under R_eval (-0.014 [-0.027, -0.002]). CER is therefore a floor check and must be read alongside style and pen.
- **Fonts are a useful recogniser prior, mostly when real data is scarce.** Fonts→real fine-tuning gives -0.004 [-0.008, -0.002] VAL CER with all writers and -0.063 [-0.080, -0.054] with 10% of the writers. Fonts alone give 0.456 and do not transfer (§7).

Table 1: Current best, word generation (VAL, 20 writers, 781 targets). CER (locator) is letter-unit and per-word-mean. R_eval CER is codepoint and corpus-level. **The two CER columns are in different units; do not compare them.** Style is M2 MRR (E_eval; chance 0.28). Pen is the generated/own-writer sub-pixel width-per-core ratio (1.00 = match). CIs are 95% writer-bootstrap.

arm	locator CER	R_eval CER all	R_eval CER ≥3cp	style M2 MRR	M2s MRR	pen raw / cal
real	0.133 [0.100, 0.167]	0.0438 [0.031, 0.059]	0.0424 [0.029, 0.057]	0.821 [0.775, 0.859]	0.768	1.00
B0	0.130 [0.103, 0.156]	0.0302 [0.025, 0.036]	0.0226 [0.016, 0.029]	0.596 [0.488, 0.711]	0.533	1.15
S t5c5	0.137 [0.101, 0.174]	0.0889 [0.065, 0.115]	0.0552 [0.038, 0.075]	**0.579** [0.521, 0.635]	0.517	1.13 / 1.06
P t5c5	0.132 [0.097, 0.171]	0.0905 [0.066, 0.118]	0.0577 [0.040, 0.079]	0.443 [0.379, 0.505]	0.416	1.15 / 1.06
B′v2 t5c5	0.151 [0.117, 0.186]	0.1412[†]	0.1024[†]	0.418 [0.355, 0.483]	0.399	1.29 / 1.07

[†]From the full R_eval rescoring table in `runs/R_eval/full_chain.log` (`Bv2_t5c5`: 0.1412 [0.1125, 0.1749] all words, 0.1024 [0.0823, 0.1272] ≥3cp); `RESULTS.md` lists only the default subset of arms. Checkpoints: S `runs/P1a3/S/`, P `runs/P1a3/P/`, B′ `runs/P1a2/B/`, B0 = BengaliScribe `models/v2`.

Source: `runs/P1a3/eval/report_tables.txt`, `runs/P1a3/eval/metrics.json`, `runs/R_eval/RESULTS.md`, `EXPERIMENTS.md` (all under `scribe_wordgen/`)

Table 2: Current best, page layout. Lower AUC = closer to real; 0.5 = indistinguishable. Plan level: 193 TEST pages (unseen texts), 5 seeds, mean ± sd. End to end: 5 writers × 2 seeds, balanced C2ST.

layout	plan line C2ST (TEST)	plan line C2ST (VAL)	plan page C2ST (TEST)	e2e line / page C2ST
manual (BengaliScribe constants)	0.980	–	1.0	0.905 / 0.999
v1 (`scribe_layout/runs/default`)	0.566 ± 0.021	0.560 ± 0.024	0.654 ± 0.043	–
v2 default (`default_unfitw_dc`)	$\mathbf{0.535 \pm 0.012}$	0.536 ± 0.027	0.590 ± 0.024	0.719 / 0.850
v2 default + writer context	–	–	–	0.619 / 0.845
null: within-page split / e2e real halves	0.496 ± 0.019	–	–	0.534 / 0.508

Source: `scribe_layout_v2/runs/compare_unfitw/compare_5seed{,_dc,_val}.txt`, `scribe_layout_v2/report/FACTS.md`, `scribe_layout_v2/runs/e2e_final/results.md`

2 Introduction and motivation

BengaliScribe (BS) turns a text into a handwritten-looking page in three steps. It reads the target writer's reference crops with an attention locator (`attn_locator`). It generates each word with a font-prior GAN conditioned on a style vector (`font_generator_pilot_sal.py`, "FG"). It then lays the words out with hand-picked constants (`write_page.py`, "WP"). We call the shipped generator **B0**. Our showcase writer is `image_2`, a page from the hcr/996 corpus that no model trains on. Its hand is thin, sparse, rounded and cursive. B0's output for it is bold, uniform and angular: a re-inked font skeleton.

2.1 Why B0 looks like a font

The planning diagnosis (`PLAN.md` §1) was checked line by line by an adversarial review (`PLAN_REVIEW.md` §0). It found these causes:

1. **The architecture copies the font.** FG is a 4.6 M-parameter U-Net over the NotoSans render (the checkpoint holds 4.587 M). It keeps the 1/4- and 1/2-resolution skips (`SKIP_SCALES=(1,1,0)`), so stroke positions reach the decoder with 2-px precision. The only module that moves ink is `StyleWarp`. A bound on its learned weights gives at most 1.38 px vertical and 5.75 px horizontal displacement on the 64×256 canvas; the architectural cap is 3.84/15.4 px. An i-kar loop of the reference writer rises 0.87 core (median) above the matra, which is out of reach. Cross-attention to the references is off (`XATTN=False`).
2. **The losses reward an averaged font skeleton.** The loss is $4\,L_1 + 1{\cdot}\text{LSGAN} + 0.3$ style-CE $+ 0.3$ CTC. The L_1 term is salience-weighted on both target and font, and the PatchGAN is conditioned on the prior. When prior and target are misaligned beyond the warp's range, the L_1 optimum is a blurred stroke along the font skeleton.
3. **Train/inference mismatch in width.** In training the prior is resized into the target's box. At inference it keeps the font's natural width, so word width and letter spacing are copied from the font.
4. **Resolution and pen are set by the data, not the writer** (Table 3). The reference writer's stroke-to-core ratio (0.053) is below BN-HTR's 10th percentile. At 64 px it is about 1.6 px wide, which the canvas cannot render sharply. `crop_ink` also stretches every target so its 5th-percentile pixel is fully black, so darkness saturates: 0.93 generated against 0.80 real.

The measured outcome on the page is a stroke of **0.39 mm generated against 0.25 mm real**, about 55% wider (`PLAN.md` §1.4, orchestrator measurement).

Table 3: Stroke width relative to letter size (skeleton EDT width ÷ body height from `measure_rows`), by data set. image_2 is the reference writer.

set	crops	width/core median (IQR)	native crop height	darkness
BN-HTR (B0's training data)	1489	0.106 (0.077–0.185)	74 px	0.98
BanglaWriting	1497	0.090 (0.066–0.135)	163 px	1.00
hcr/996 (60 pages)	1447	0.075 (0.060–0.103)	102 px	0.82
hcr image_2 (reference writer)	57	**0.053** (0.048–0.067)	107 px	0.79
generated for image_2 (64-px canvas)	60	0.063 (0.045–0.076)	64	0.97

Source: `scribe_wordgen/PLAN.md` §1.4 (script `scratch/pen_ratio.py`; re-run identically in `PLAN_REVIEW.md` §0)

Per-writer BN-HTR medians span 0.073–0.186 (p10–p90 over 110 writers). The second problem is **layout**. WP places words with uniform random gaps, a tiny fixed line slope, and a fixed pitch of $2.7\times$ body height, so every line is straight and evenly spaced.

2.2 Goal and overall design

The goal is a page generator that *writes like the reference writer*: the writer's letter construction and pen, laid out the way real people lay out lines. Figure 1 shows the pieces built in this project and how they are evaluated. Each evaluator is independent of the generator it scores: none is ever used as a training loss.

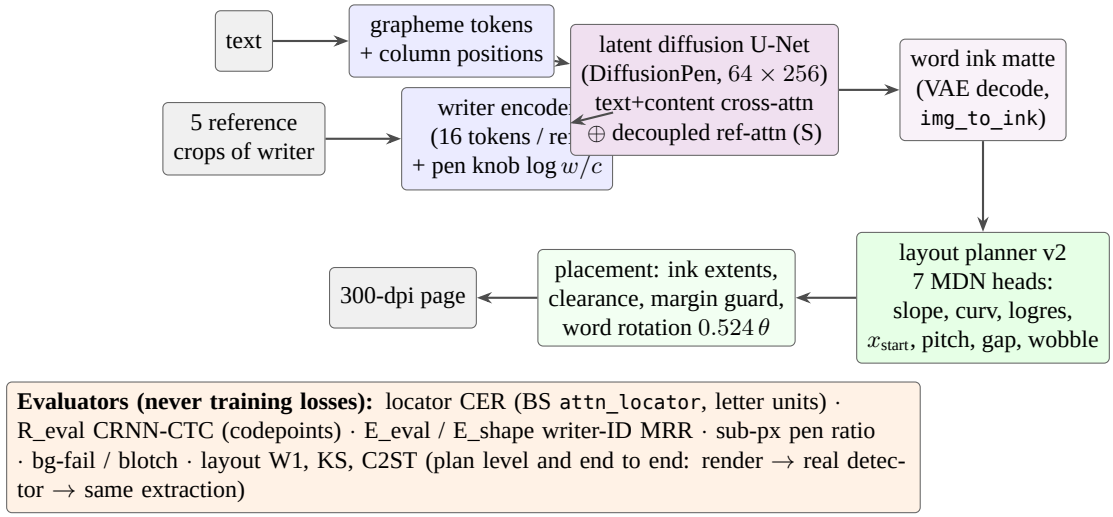


Figure 1: The system. Upper row: the word generator (Arm S configuration; Arm P lacks the reference-token attention). Lower row: the learned layout and its placement rules. Bottom: the evaluators.

3 Data and protocol

3.1 Data sets

Table 4: Data sources. Only BN-HTR and BanglaWriting are used to train word models. hcr/996 is used for layout and as an evaluation-only showcase (image_2).

source	word crops	writers	native size	notes
BN-HTR	92,380 labelled + filtered (from 108,181)	149	median crop h 74 px, core ≈28–30 px	93,758 labelled and ≤16 graphemes, minus 1,378 flagged by `label_filter.json`; folder = writer
BanglaWriting (BW)	20,523	260 (`style_id` = page = person)	median crop h 163 px, core ≈49 px	one page per writer; manual labels
hcr/996	≈48.7k word pieces	unknown	A4, 300 dpi, core median 41 px	27 texts copied by many writers; 963 usable pages for layout; no writer IDs

Source: `scribe_wordgen/PLAN.md` §1.6, §2.1; `runs/T0/REPORT.md`; `scribe_layout/report/scribe_layout_report.tex` §2

3.2 Writer-disjoint splits

Splits are fixed in `data_cache/splits.json`, and a test asserts that no writer appears in two of them (`runs/T0/REPORT.md`):

- **BN-HTR:** TEST = {15, 22, 24, 28, 30, 48, 51, 54, 69, 136} (5,768 crops); VAL = {99, 117, 127, 142, 149}; TRAIN = the remaining 134 writers. TEST writers are in BS's generator and locator hold-out, so B0 is scored

on writers it never saw. The VAL writers are in the TRAIN set of BS's style encoder `ms4`; this was verified, not assumed (`PLAN_REVIEW.md` §0).

- **BanglaWriting:** `style_ids` hash-sorted (SHA-256) into 220 TRAIN / 15 VAL / 25 TEST. `dp_words_b64` (baseline B3) saw all 260 BW writers, so B3 is fair only on BN-HTR.
- **hcr/996 (layout):** split *by source text*: 18 train / 4 val / 5 test texts (test texts {4, 6, 11, 15, 18}), giving 614 / 144 / 193 pages. The v1 report claimed test lines are by unseen writers. The v2 review notes that writers likely repeat across texts, which cannot be checked without writer IDs (§8.1).

3.3 The VAL protocol (word generation)

Unless stated otherwise every word-generation number uses this protocol (`EXPERIMENTS.md`, header):

- **20 VAL writers**: the 5 BN-HTR VAL writers plus the 15 BW VAL writers, with **781 target words** in total.
- **5 conditioning references per writer.** For BN-HTR these come from page A and the targets from other pages. For BW the refs and targets come from the same (only) page, which inflates BW style metrics through page cues (`runs/P1a/REVIEW.md` §4).
- Targets whose text is among the refs are excluded. Punctuation-only targets (54/781, all BW) are excluded from the headline CER.
- **Seeds.** B0 and real have one sample per word. Diffusion arms have two seeds from eval_v3 onward. Paired comparisons merge on (dataset, writer, idx, seed_k) and never zip lists.

3.4 TEST status

No generator has been evaluated on TEST. The final TEST evaluation (Checkpoint C) has not been run. The only TEST contact is that the independent recogniser R_eval was scored once on real TEST crops (7,665 words, §7). For layout, all design choices were compared on TEST (selection on test). A selection-free VAL check was run afterwards (§5.4).

3.5 Data-handling facts that affect the numbers

- **The BanglaWriting word-crop PNGs are missing from disk.** The files under `Diffusion_model_dataset_building/dataset/word_crops/` that `index_bw.csv` points to were deleted. The script `wordgen/recog/bw_regen.py` rebuilds the crops in memory from the LabelMe page JSON and JPG in `banglawriting/converted/`. It is a faithful port of the original `parse_json`/`crop_box` and label-checks every crop.
- **R_eval BW coverage.** The R_eval real-crop cache holds **20,482 of the 20,523** BW words (17,539 TRAIN + 1,046 VAL + 1,897 TEST rows in `runs/R_eval/cache/bw_*.jsonl`). The 41 missing words are the rows that already failed `measure_rows` in the T0 data index (`runs/T0/REPORT.md` §4.2: ok 20,482 of 20,523); the cache's per-split counts equal T0's ok-row counts.
- **Core-normalised canvas: clip target not met.** The plan's 160-px core-normalised canvas (core 32 px) clipped 22.6% of BN-HTR and 21.1% of BW TRAIN crops, almost all at the top (`runs/T0/REPORT.md` §4.3). A writer-prior fallback cut this to 18.98% / 14.57% (`runs/T0b/REPORT.md`), still far from the $<1\%$ target. The SD-VAE round trip also fails the plan's IoU gate (≥ 0.93) at core 32 (image_2 0.862; `PLAN_REVIEW.md` B1). All generators therefore stayed on DiffusionPen's height-normalised 64×256 `fit_canvas`. The learned layout rescales words by measured core anyway.

4 Evaluation methodology

Every word metric is computed per writer and then averaged over writers. The CI is a writer-level bootstrap: 2,000 resamples, 95% percentile interval. A "tie" means the paired CI includes 0.

4.1 Locator CER (letter units, per-word mean)

BS's frozen `attn_locator` reads each word on a fixed 64×256 canvas. The ink is tight-cropped and left-aligned; wider words are resized to fit (eval_v3 truncated B0 words wider than 256 px, which affects about 10% of B0). For

target y_i and reading $\hat{y}_i$,

$$\mathrm{CER}_i^{\mathrm{loc}} = \min\Big(1,\ \frac{d_{\mathrm{edit}}(\hat{y}_i, y_i)}{|y_i|}\Big), \qquad \mathrm{CER}_w^{\mathrm{loc}} = \frac{1}{|T_w|S}\sum_{i\in T_w}\sum_{s}\mathrm{CER}_{i,s}^{\mathrm{loc}}, \qquad \mathrm{CER}^{\mathrm{loc}} = \frac{1}{20}\sum_{w}\mathrm{CER}_w^{\mathrm{loc}}. \tag{1}$$

The edit distance is in the locator's letter units. The cap at 1 matters: without it, garbage reads of 1-character punctuation targets gave CERs of 23–29 and 15% of words had CER > 1 (`runs/P1a/REVIEW.md` §1). The locator is BN-HTR-trained. Its own real-crop CER is 0.056 on BN-HTR VAL and 0.236 on BW VAL (`runs/R_eval/RESULTS.md`), so it separates arms poorly on BW.

4.2 R_eval CER (codepoints, corpus level)

R_eval is an independent CRNN-CTC recogniser (§7). Its CER is micro-averaged over Unicode codepoints and not capped:

$$\mathrm{CER}^{R} = \frac{\sum_i d_{\mathrm{edit}}(\hat{y}_i, y_i)}{\sum_i \max(|y_i|, 1)}, \tag{2}$$

where $|\cdot|$ counts codepoints. The CI resamples writers and recomputes the corpus ratio. A paired difference $\mathrm{CER}^R(a) - \mathrm{CER}^R(b)$ is computed on the same rows (same gold, same writer) and bootstrapped the same way (`wordgen/recog/metrics.py`). Results are reported for all words and for words of $\geq$3 codepoints. **Locator CER and R_eval CER differ in unit (letters vs codepoints) and in aggregation (capped per-word mean vs corpus ratio). They must never be compared with each other.**

4.3 Style: writer-identification MRR (M2, M2s)

E_eval and E_shape are ResNet-18 writer-embedding networks trained with SupCon plus cosine-softmax CE, on TRAIN writers only. **E_eval** sees the normalised ink matte (construction *and* pen). **E_shape** sees a spur-pruned skeleton redrawn at a fixed thin width (construction only). Every source goes through one preprocessing path that ends at the generators' 64×256 canvas (`wordgen/eval/writer_id.py`). For a query word q of writer w, each candidate writer v of the same dataset has a gallery vector g_v: the L2-normalised mean embedding of real crops disjoint from the refs and the query. The galleries are cross-fitted by index parity, and crops with the query's text are removed. With r_q the rank of w among candidates by $\cos(e_q, g_v)$:

$$\mathrm{MRR} = \frac{1}{20}\sum_{w}\frac{1}{|Q_w|}\sum_{q\in Q_w}\frac{1}{r_q}, \qquad \mathrm{margin}_q = \cos(e_q, g_w) - \max_{v\neq w}\cos(e_q, g_v). \tag{3}$$

Candidates are the VAL writers of the query's own dataset: 5 for BN-HTR (chance MRR 0.457) and 15 for BW (0.221). Pooled over 20 writers this gives chance top-1 0.10 and chance MRR 0.28. CIs come from a hierarchical bootstrap (writers within dataset, then queries within writer), paired across methods (`wordgen/eval/style_metric.py`).

Validity (`runs/style_eval/tables.md`). Real crops score top-1 0.723 [0.651, 0.786] and MRR 0.821 [0.775, 0.859]. Font renders sit near chance (NotoSans MRR 0.289, Atma 0.268). B0 is far above fonts (B0$-$Noto $+0.307$ [+0.156, +0.464]), so the metric sees writer-conditioned shape, not just "handwriting vs font".

4.4 Pen metric (sub-pixel width per core)

Per word (`wordgen/eval/pen.py`): $M = \{\text{ink} > \text{Otsu}\}$, core c from `measure_rows`, skeleton K, and

$$w_{\mathrm{sub}} = \frac{\sum_{p:\,\mathrm{ink}_p > 0.17}\mathrm{ink}_p}{|K|}, \qquad \rho = \frac{w_{\mathrm{sub}}}{c}, \qquad \mathrm{pen}_w = \frac{\mathrm{median}_{i\in\mathrm{gen},w}\,\rho_i}{\mathrm{median}_{i\in\mathrm{real},w}\,\rho_i}. \tag{4}$$

Ink mass scales with s^2 and skeleton length and core with s, so ρ is almost resolution-invariant. The reported value is the median of pen_w over writers, with p10–p90 and a bootstrap CI of the median. This replaced eval_v2's integer EDT width. On a 64-px canvas that metric takes only the values $\{1, 1.83, 3, 3.47, \dots\}$ and reported 1.8–2.2$\times$ for every generator, where like-for-like measurement gives 1.13–1.36$\times$ (`runs/P1a2/REVIEW.md` §1).

4.5 Background metrics

bg-fail: a word fails when more than 60% of its ink bounding box is ink, or the box's 25th-percentile ink exceeds 0.1. The rate is a mean of per-writer rates. **Faint blotch**: a non-punctuation word with >2% of the canvas covered by faint ink (0.17–0.5) more than 2 px from strong ink. Writer bnhtr_117, a grey-paper scanner, is tracked separately.

4.6 Statistics for claims

All decisions are made on VAL. A claim of "A beats B" needs the paired CI of A−B to exclude 0. The paired bootstrap resamples the same writers (and, for style, the same query positions) for both arms. Each table also gives the count of writers favouring the first arm.

5 Page layout: v1 to v2

5.1 What is measured

Layout is learned from real ink geometry on hcr/996. A word detector (the project's `unified_cascade` engine: a Swin-T Mask R-CNN line detector and a word detector on line strips) gives 50-point word polygons. Only the word polygons are kept, because the line polygons merge on sloped pages. Lines are then rebuilt by chaining words with a robust quadratic fit. Ink is binarised (Otsu capped at grey 170, components $<$30 px removed) and assigned to polygons. All geometry is in a per-page writer unit **U** = the median word core height (80th minus 20th percentile of ink rows), about 3.5 mm and 31–54 px at 300 dpi. U was chosen because the pitch/U ratio is most stable across writers (CV 0.189).

Real layout statistics (train split; v1 report Table 4). The real-data statistics shape the model. Gaps have median 1.14 U and lag-1 Spearman 0.49 within a line. Gaps tighten along the line (-0.13) and their line mean is a writer habit (ICC 0.37). Slope is a strong writer habit (ICC 0.77) and persists to the next line ($r = 0.87$). Curvature is weak (ICC 0.17). Pitch has median 4.40 U (ICC 0.35). The residual (wobble) scale has median 0.098 U. So spacing and slope cannot be i.i.d., and the generator must condition on the page's history.

5.2 The model and its learned parameters

`LayoutModel` has seven small 1-D mixture-density heads ($k = 4$ components, hidden 64), trained by exact NLL with teacher forcing, full batch, Adam (lr 10^{-3}, wd 10^{-5}), patience 30. A page is generated line by line by the chain rule. For line k with page-history vector H_k (20 running statistics of previous lines):

$$p(\text{line}_k \mid H_k) = p(c_1 \mid H)\, p(c_2 \mid H, c_1)\, p(\log s \mid H, c_1, c_2)\, p(x_{\text{start}} \mid \cdot)\, p(\text{pitch} \mid \cdot) \times \prod_j p(\text{gap}_j \mid w_j, w_{j+1}, \text{gap}_{<j}, \cdot) \prod_j p(r_j \mid \log s, j/n). \quad (5)$$

The headline trajectory is $y = c_0 + c_1(x - x_m) + c_2(x - x_m)^2$, centred on the line's x midpoint, plus per-word residuals r_j ("wobble") with robust scale s. Curvature needs $\geq$5 words and slope $\geq$3. The model learns slope, curvature, residual scale, line start, pitch, gaps (autoregressive) and wobble. v2 adds fitted constants and placement rules. Table 5 lists all of them.

Table 5: Learned and fitted layout parameters of the final default (v2 `default_unfitw_dc`). All constants are fitted on the train split.

parameter	value	role
MDN heads	7 heads, $k = 4$, hidden 64; best epoch 152 (stopped 183)	slope, curv, logres, x_{start}, pitch, gap, wobble
training rows	slope 4,942; curv/logres 4,050; x_{start} 4,814; pitch 4,258; gap 25,546; wobble 26,878	per-head targets
val NLL (nats)	slope -2.363, curv -1.004, logres 1.071, x_{start} 0.965, pitch 1.152, gap 0.669, wobble -0.606	
combined word anchor	$w = 0.349$ headline + 0.651 median-ink-row	inverse-noise weighting
anchor noise	true wobble var 0.00906 U^2; noise var headline 0.01176, mid 0.00631, combined 0.00411	deconvolution of wobble
wobble de-noise factor	$\sqrt{1 - \text{share}_{\text{combined}}} \approx 0.83$ (share 0.29–0.33)	generation-time scale
unfit targets	per-word leverage inflation $1/\sqrt{1-h_i}$, cap 4; line scale keeps the median-MAD table	restores physical wobble
curvature deconvolution	noise = 38.4% of curv target variance; MDN likelihood $\sum_k \pi_k \mathcal{N}(c; \mu_k, \sigma_k^2 + v_i)$	sampling is noise-free
line clearance	min distance $\geq$ 2.603 U, min/mean $\geq$ 0.797 (train 1st pct.); 30 redraws, else push down	stops line collisions
word rotation	$\beta = 0.524\times$ line tangent angle (Huber fit, $n = 26{,}653$)	tilts words with the line
placement	ink-extent edges (`INK_LEVEL` 0.35); planner reset per page; margin guard ± 0.3 U	BS integration

Source: `scribe_layout_v2/runs/default_unfitw_dc/metrics.json`, `.../default_unfitw_dc/tilt.json`, `scribe_layout_v2/runs/anchor_noise.json`, `.../line_clearance.json`, `scribe_layout_v2/report/FACTS.md` items 1–11

5.3 Metrics: W1, KS and C2ST

Every generator receives each real TEST page's word widths and line breaks. Its plan is converted to raw coordinates and *re-measured by the same code* that measured the real lines. For each marginal quantity q (gap, slope, curvature, residual scale, drift $y_n - y_1$, pitch, x_{start}) we report the Wasserstein-1 distance and the KS statistic between generated and real distributions:

$$W_1(P, Q) = \int |F_P(t) - F_Q(t)|\, dt, \qquad \text{KS}(P, Q) = \sup_t |F_P(t) - F_Q(t)|. \tag{6}$$

The **classifier two-sample test (C2ST)** trains a classifier (logistic regression unless marked MLP) to tell real lines (or pages) from generated ones. It uses grouped 5-fold cross-validation, where each real page and its generated copy form one group, and reports out-of-fold AUC. AUC $= 0.5$ means indistinguishable. Line features (lines with $n \geq 5$): mean and sd of gaps, gap lag-1, slope, curvature, residual scale, drift, gap–width correlation. Page features: mean and sd of line-mean gaps, mean and sd of slope, slope lag-1, mean curvature, mean residual scale, mean pitch.

Nulls. The "shuffled real" baseline (another page's geometry) is **not** a floor. It breaks the gap–width relation, which is a C2ST feature (0.018 vs real 0.130). The comparable null splits each real page into two labels inside one group: 0.496 ± 0.019. End to end, the floor is real TEST pages split in half: line 0.534, page 0.508.

5.4 Results in chronological order

v1 (`scribe_layout/`). Table 6 is v1's own ablation on its own harness. The learned model reduces gap W1 from 0.490 (naive) and 0.559 (manual) to 0.038. Vertical structure (slope, drift, pitch) comes from the vertical heads, spacing from the gap head; each alone leaves the other at manual's level. The weak point is x_{start} (W1 1.31 U), worse than shuffled real (0.43). Model selection chose $k = 4$ (summed val NLL 0.417, against 1.710 for $k = 1$; `scribe_layout/runs/model_ablation.json`).

Table 6: v1 ablation (v1 harness, TEST, 5 seeds, mean ± sd). W1 is in U.

method	gap W1	slope W1	drift W1	pitch W1	x_{start} W1	line C2ST
naive	0.490	0.038	1.217	0.849	1.811	1.000
manual	0.559	0.036	1.143	1.607	1.713	0.977
learned (v1)	**0.038 ± 0.007**	0.005	0.158 ± 0.030	0.092 ± 0.015	1.314 ± 0.021	0.619 ± 0.011
learned, spacing only	0.043	0.036	1.138	1.605	1.318	0.874
learned, vertical only	0.561	0.005	0.159	0.085	1.713	0.944
shuffled real	0.106 ± 0.034	0.006	0.187	0.194	0.434 ± 0.104	0.588

Source: `scribe_layout/runs/eval/ablation.md`

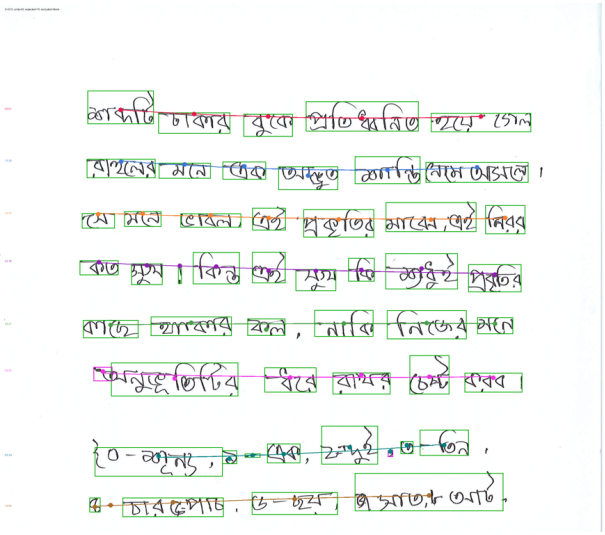


Figure 2: Extracted geometry on a real page (v1 report): word anchors, fitted line trajectories and residuals.
Source: `scribe_layout/report/fig/image_103_geom.png`

v2 harness and the move to "unfit" targets. v2 (`scribe_layout_v2/`) re-evaluates v1 on a stricter harness with more C2ST features: there, v1's line C2ST is 0.566 ± 0.021. Table 7 gives the sequence of v2 models. The first v2 models (`v2_history`, `v2_latent`) learned the *fitted* wobble. Fitting a quadratic to n points shrinks the residuals ("fit shrinkage"), and replaying them adds a second fit at measurement. Training on inflated ("unfit") targets lowered the line C2ST from 0.638 to 0.539 (single seed then 5 seeds).

A bug in the unfit targets, and the fix. A review (M2) found that the inflation table matched the *line* robust scale but was also applied to every per-word residual. This made per-word wobble overshoot: in a simulation with i.i.d. wobble of sd 0.150, the old table recovers 0.311 at $n = 5$, while per-word leverage correction recovers 0.142. The 2-D FFT C2ST had already flagged this. Scaling the drawn wobble moved it from 0.683 ($\times 1.0$) to 0.578 ($\times 0.5$) (3 seeds). The leverage fix `unfitw` gave the best line C2ST, 0.521 ± 0.019.

Curvature deconvolution (the default). In end-to-end runs the real detector split 89 of 462 drawn v2 lines (19%), against 6 of 355 (1.7%) for manual. At the split points, adjacent words stepped by a median 1.2 U, more than the chainer's 1.0 U join threshold. The cause was lines that bow too much. Curvature is sampled before the word count is known, and fitted c_2 on 5–6-word lines is mostly noise (measured sd 0.00129, noise 0.00092, implied true 0.00091). An extreme-deconvolution likelihood removes this: sampled sag matches real ($|\text{sag}| > 0.5$ U on 9.3% of lines, real 10.8%, `unfitw` 17.5%), and end-to-end splitting halves to 52/469 (11.1%). Its C2ST is within noise of `unfitw`. `unfitw_dc` was chosen for realism, not for C2ST.

A rejected variant. A Huber-aware per-word correction (`unfitwh_dc`) is exact for i.i.d. wobble, but was worse on all three C2STs on both splits (6/6 comparisons). Real wobble is smoother than i.i.d. (lag-1 -0.38 vs -0.48), so the larger edge inflation overshoots.

Table 7: Plan-level C2ST AUC by model, TEST (193 pages), 5 seeds, mean ± sd. Resid-scale W1 is given by words per line.

model	line C2ST	page C2ST	2-D FFT C2ST	resid_scale W1, n=5 / 6–7 / 8–10 / 11+
manual (BengaliScribe)	0.980	1.0	0.99	–
v1	0.566 ± 0.021	0.654 ± 0.043	0.567 ± 0.034	.011 / .025 / .055 / .074
v2_history (1 seed)	0.638	0.752	–	–
v2_history_unfit (old table)	0.539 ± 0.016	0.660 ± 0.004	0.653 ± 0.024	.039 / .028 / .029 / .058
v2 unfitw (leverage)	$\mathbf{0.521 \pm 0.019}$	$\mathbf{0.581 \pm 0.026}$	0.566 ± 0.025	.009 / .011 / .041 / .062
v2 unfitw_dc (default)	0.535 ± 0.012	0.590 ± 0.024	0.550 ± 0.020	.007 / .011 / .042 / .061
v2 unfitwh_dc (Huber, rejected)	0.544 ± 0.012	0.614 ± 0.011	0.583 ± 0.029	.009 / .014 / .043 / .068
shuffled real (not a floor)	0.584 ± 0.016	0.605 ± 0.012	0.529 ± 0.050	.021 / .018 / .054 / .100
within-page null	0.496 ± 0.019	–	–	–

Source: `scribe_layout_v2/runs/compare_unfitw/compare_5seed.txt`, `compare_5seed_dc.txt`, `compare_5seed_wh_test.txt`; v2_history row: `scribe_layout_v2/runs/eval/metrics.json`; null and manual: `scribe_layout_v2/report/FACTS.md`

Selection-free VAL check. Every choice above was made on TEST, so the TEST numbers are optimistic. Table 8 repeats the comparison on VAL. The leverage-family models beat v1 on line and page C2ST on VAL too. The old-table `unfit` model, which looked better than v1 on TEST (0.539), is not better on VAL (0.566 vs 0.560): that earlier claim was a selection effect.

Table 8: Plan-level C2ST on the VAL split (4 texts), 5 seeds.

model	line C2ST	page C2ST	FFT C2ST	resid_scale W1 by n
v1	0.560 ± 0.024	0.631 ± 0.040	0.531 ± 0.033	.013 / .025 / .055 / .089
v2 unfit (old table)	0.566 ± 0.028	0.660 ± 0.022	0.599 ± 0.051	.049 / .030 / .024 / .050
v2 unfitw	0.527 ± 0.030	0.575 ± 0.039	0.528 ± 0.024	.012 / .011 / .035 / .072
v2 unfitw_dc (default)	0.536 ± 0.027	0.593 ± 0.031	0.535 ± 0.044	.011 / .011 / .035 / .069
v2 unfitwh_dc	0.551 ± 0.022	0.619 ± 0.043	0.551 ± 0.049	–
shuffled real	0.577 ± 0.023	0.597 ± 0.030	0.480 ± 0.034	.016 / .017 / .055 / .089

Source: `scribe_layout_v2/runs/compare_unfitw/compare_5seed_val.txt`, `compare_5seed_wh_val.txt`

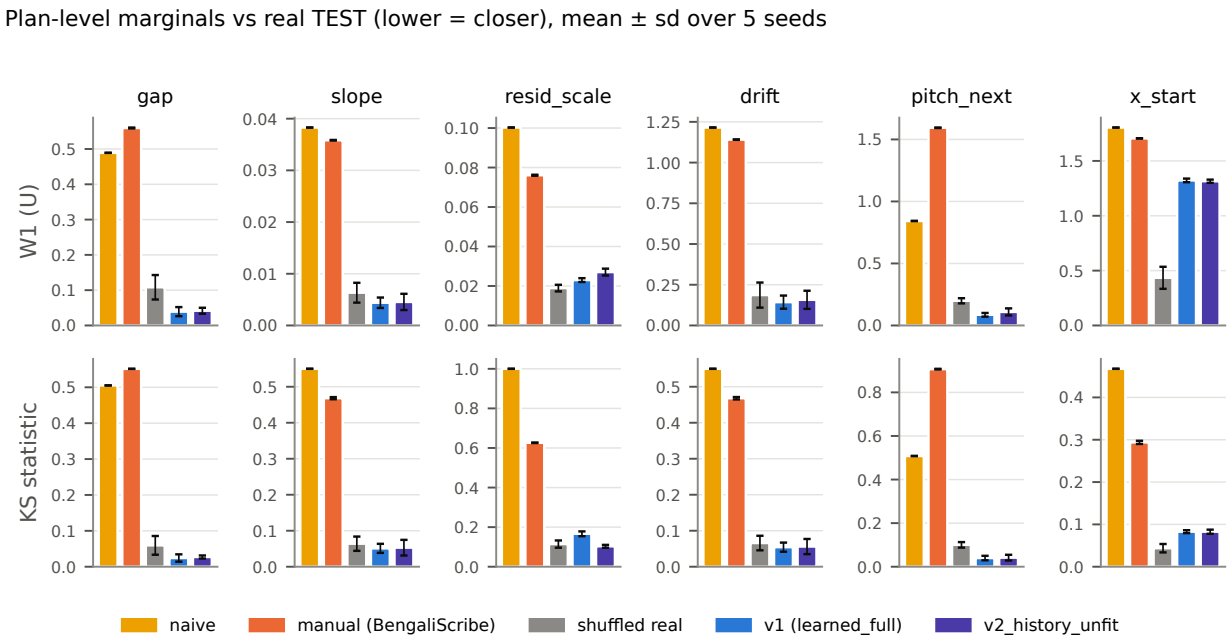


Figure 3: Plan-level marginals against real TEST on the v2 harness: W1 (top) and KS (bottom). Learned layouts (blue, violet) match gaps, slope, drift and pitch far better than naive or manual layout. x_{start} remains the weakest learned quantity; shuffled real does better there.

Source: `scribe_layout_v2/runs/eval/metrics.json` (via `make_figs.py`)

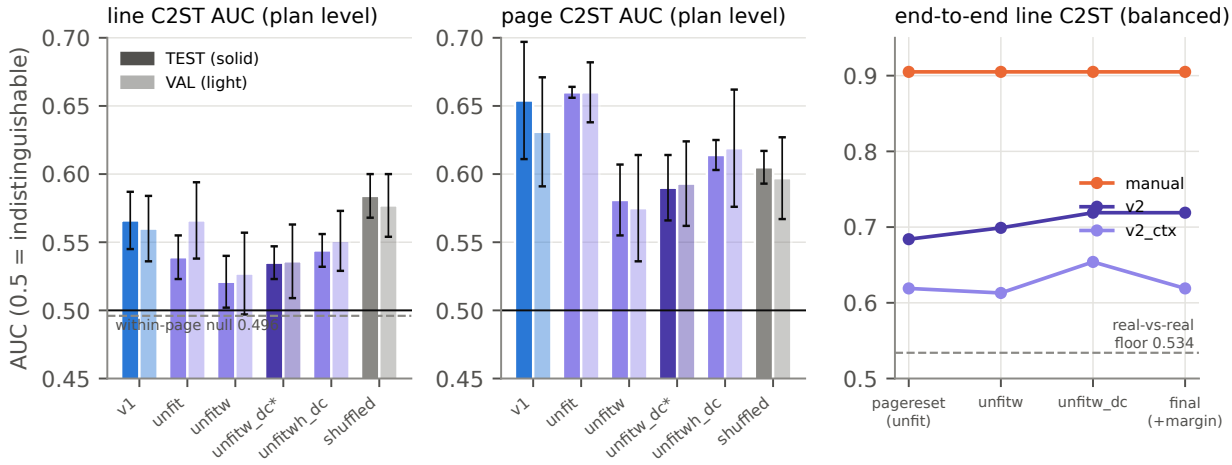


Figure 4: C2ST AUC per model. Left and centre: plan level, TEST (solid) and VAL (light), 5 seeds, ± sd. "*" marks the default. Right: end-to-end line C2ST, balanced protocol only. The two earliest end-to-end runs used a biased C2ST and are excluded (see text).

Source: `scribe_layout_v2/runs/compare_unfitw/compare_5seed*.txt`, `scribe_layout_v2/runs/e2e_*/results.md`

End-to-end evaluation. BengaliScribe renders pages for 5 reference writers × 2 seeds. The real detector and the same extraction then measure them, and they are compared with the 193 real TEST pages. Variants: `manual`, `v2` and `v2_ctx` (conditioned on the writer's own real geometry). **The first two runs (`e2e_unfit`, `e2e_unfit_ink`) used a biased C2ST.** Pooled out-of-fold AUC with one label per page group put the real-vs-real "floor" at 0.34 (line) and 0.115 (page). Their C2ST values are not comparable with later runs; only their W1 columns are shown. From `e2e_pagereset` onward the C2ST is balanced (folds dealt per label; mean of per-fold AUCs), with floor 0.534 / 0.508.

Table 9: End-to-end results (W1 in U; AUC). ‡Biased C2ST: not comparable. Recovered = extracted words / drawn words.

run (model)	method	gap	slope	drift	pitch	x_{start}	line / page C2ST	recovered
e2e_unfit (canvas-width)	v2	0.602	0.015	0.531	0.716	3.837	0.750 / 0.901‡	0.969
	v2_ctx	0.381	0.013	0.427	0.924	4.416	0.595 / 0.901‡	0.910
e2e_unfit_ink (ink extents)	v2	0.471	0.016	0.503	0.712	4.037	0.667 / 0.840‡	0.952
	v2_ctx	0.207	0.010	0.343	1.001	4.438	0.532 / 0.918‡	0.866
all balanced runs	manual	0.685	0.035	0.965	1.839	0.948	0.905 / 0.999	1.004
e2e_pagereset (unfit)	v2	0.404	0.013	0.396	0.670	4.065	0.684 / 0.834	0.965
	v2_ctx	0.267	0.017	0.558	0.770	3.783	0.619 / 0.889	0.926
e2e_unfitw	v2	0.376	0.012	0.343	0.346	2.278	0.699 / 0.851	0.960
	v2_ctx	0.197	0.019	0.610	0.480	2.703	0.613 / 0.842	0.933
e2e_unfitw_dc	v2	0.387	0.021	0.645	0.465	1.453	0.719 / 0.844	0.983
	v2_ctx	0.183	0.023	0.695	0.535	2.494	0.654 / 0.847	0.929
e2e_final (default + margin guard)	v2	0.425	0.021	0.672	0.390	1.239	0.719 / 0.850	0.981
	v2_ctx	0.198	0.020	0.621	0.503	2.130	0.619 / 0.845	0.940

Source: `scribe_layout_v2/runs/e2e_{unfit,unfit_ink,pagereset,unfitw,unfitw_dc,final}/results.md`

Reading the end-to-end numbers:

- **The planner reset was a real fix.** The history used to run across pages, reaching 56–72 lines against at most 16 in training. Slopes random-walked to the -0.2 clamp and words were drawn past the page bottom. After the reset, page slope means stay near the writer's (e.g. image_104 with context: $-0.053 \cdots -0.009$ against real -0.023).
- **Slope and drift W1 are not comparable across runs with different split rates** (survivor bias). Split lines are the steep and bowed ones. In `e2e_unfitw`, kept lines had median |slope| 0.026 against 0.046 for split lines. All drawn `dc` lines have median |slope| 0.027 (q90 0.071), against real 0.029 (0.086).
- **Placement bias.** Ink extents use `INK_LEVEL` 0.35 while page Otsu is ≈0.39, which leaves a $+0.067$ U bias in every measured gap.

- **Line splitting.** With 57 of 453 drawn lines split (12.6%) in `e2e_final`, end-to-end metrics see only the surviving lines, and the large x_{start} W1 mostly comes from fragments that start mid-page.

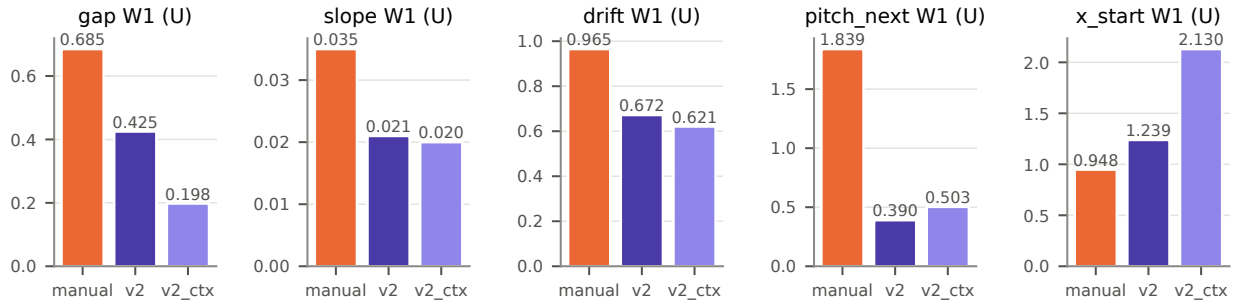


Figure 5: End-to-end W1 to real TEST for the final run (`e2e_final`).
Source: `scribe_layout_v2/runs/e2e_final/results.md`

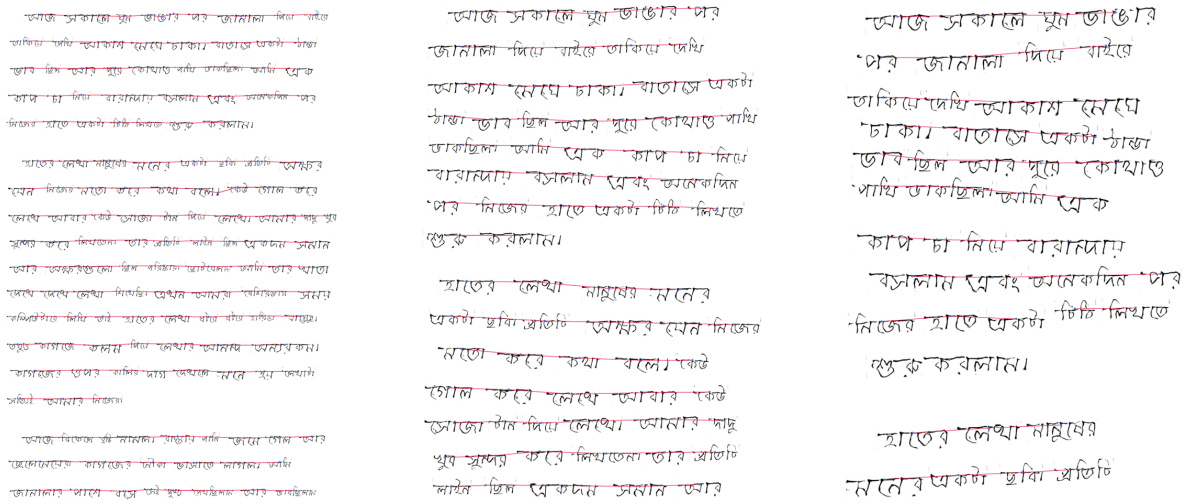


Figure 6: `e2e_final` pages for reference writer image_104 (BengaliScribe B0 words): manual layout (left), v2 (centre), v2 with writer context (right).
Source: `scribe_layout_v2/runs/e2e_final/thumbs/image_104_{manual,v2,v2_ctx}.png`

Final default model. `scribe_layout_v2/runs/default_unfitw_dc/model.pt` is the default of `bscribe_run.py` and `relayout.py`. It runs with the margin guard: a line whose ink would cross the bottom margin starts a new page and is replanned from scratch. A test asserts that ink stays within the margins ± 0.5 U; it fails on the old code. The layout test suite grew from 32 tests (≈8 min) to 70 (≈17 s). The latent writer-style variant is **not** the default: a context bug fix (causal-prefix z) improved the context case but regressed the no-context model (page C2ST 0.684 → 0.829), and the issue is unresolved.

5.5 Spectral side-studies, with corrections

2-D FFT layout-realism metric (`scribe_layout_fft2d/`). A layout map is the filled word boxes at 8 px/U, Hann-windowed. Its 2-D power spectrum is summarised as radial PSD (32 bins), angular energy (18 bins) and the vertical line-pitch peak. **Only the controlled (text-matched) comparison is valid.** The uncontrolled version is confounded: single non-layout features alone separate real from generated (text-block height AUC 0.985, median word width 0.90–0.93), so its AUCs of 0.96–1.0 measured text, page fill and glyphs, not layout. In the controlled test every generator runs on each real page's own word widths, both sides use the same fixed box, and confound AUCs are 0.5 by construction.

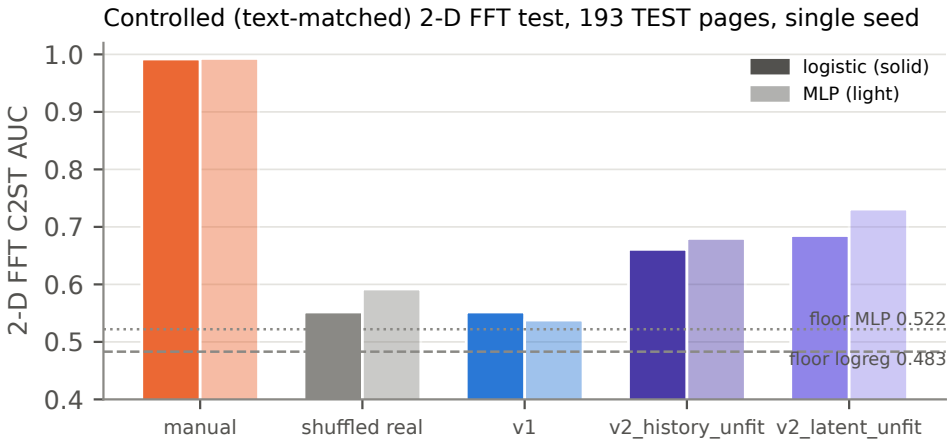


Figure 7: Controlled 2-D FFT C2ST, single seed. Single-seed AUCs move by up to ± 0.09; the 5-seed FFT C2STs are in Table 7. Pitch ablations on real pages show that the spectrum is dominated by line-pitch regularity. Shuffling pitch moves the mean log-PSD $6\times$ more (L2 0.080) than zeroing slope (0.013) or removing wobble (0.013).

Source: `scribe_layout_fft2d/runs/results.md` (controlled section only), `scribe_layout_v2/report/FACTS.md`

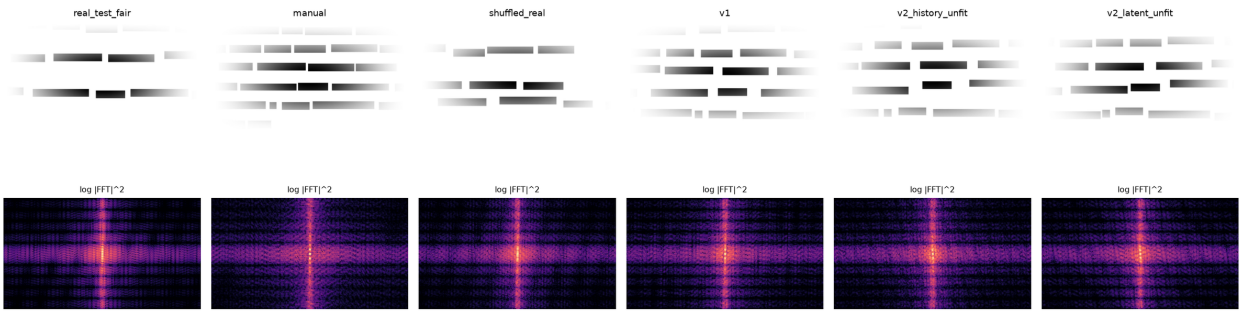


Figure 8: Controlled layout maps (fixed word boxes) used by the 2-D FFT metric: real page and generated layouts on the same word widths.

Source: `scribe_layout_v2/report/figs/example_layout_maps_fair.png`

1-D Fourier model of the headline (`scribe_layout_fourier/`). Dense headline traces of real lines follow a power law, PSD $\propto f^{-\alpha}$ with $\alpha = 1.06$ ($R^2 = 0.99$, 5,099 train lines), with no word-period peak. A spectral-synthesis generator samples Gaussian-process headlines with this spectrum. Its `runs/results.md` states that Fourier is "17$\times$ better" than v1 (log-spectral RMSE 3.98 vs 0.23 nats). **That claim is wrong.** The 0.23 belongs to a continuous trace that no layout of rigid words can draw. v1's 3.98 is a floor artefact, because the comparison ran on traces that a layout cannot realise. The corrected, fair comparison uses layout-realisable traces (rigid, rotated words) and the band with wavelength $\geq$4 U. The distance to real is then: manual 3.786, **v1 0.702, Fourier 0.668**, continuous Fourier (not realisable) 0.244. **Fourier and v1 are essentially equal.** The "6$\times$ better x_{start}" claim from the same study came from including indents.

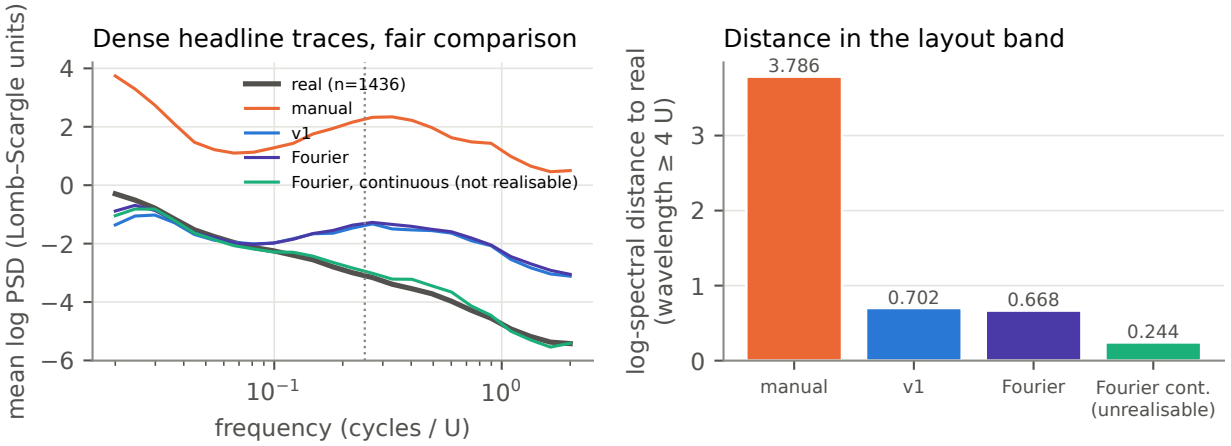


Figure 9: Corrected Fourier comparison. Left: mean log PSD of dense headline traces (dotted line: the 4 U wavelength limit). Right: log-spectral distance to real in the layout band.

Source: `scribe_layout_fourier/runs/fair_dense_psd.json`; correction in `scribe_layout_v2/report/FACTS.md`

6 Word generation

6.1 From diagnosis to design

The diagnosis (§2) implies that a higher-resolution B0 would still copy the font: construction has to be learned from the references by a model that is not tied to a font raster. The primary approach is a **few-shot latent diffusion word generator** built from DiffusionPen: an SD-1.5 VAE, a U-Net with CANINE text cross-attention, and a style vector from 5 references. Candidates are re-adapted from the IAM checkpoint on TRAIN writers only. `dp_words_b64`, which saw every BW writer, is used only as a probe and a baseline (B3). Table 10 lists the arms.

Table 10: Generator arms.

arm	definition
B0	BengaliScribe font-prior GAN (NotoSans prior), `models/v2`
B3	`dp_words_b64`: DiffusionPen fine-tuned on all 260 BW writers (fair only on BN-HTR)
A, B (P1a)	DiffusionPen from IAM on TRAIN, 40 crops/writer cap, 20k steps; B adds an 8-ch painted grapheme content map
A′, B′ (P1a′)	full TRAIN (98,856 crops, 354 writers), white-paper normalisation, 40k steps; B′ adds grapheme tokens + column-position encodings in cross-attention
B′v2	B′ sampled with `fit_canvas_white2` (bgnorm2) references, no retraining
P	B′ + pen knob (scalar embedding of the refs' log width/core) + learned null style + rebalanced sampling; 16k steps
S	P + 16 reference tokens per ref (Bengali SupCon ResNet-18) in decoupled (IP-Adapter-style) cross-attention; 24k steps
_tXcY / _cal	text guidance X and content guidance Y; _cal = page-scale pen calibration (shrink mode)

6.2 probe_day1: a visible first probe

`dp_words_b64` was sampled on image_2's text with three reference sets, and fine-tuned for 120 steps on image_2's confident crops. The result was qualitatively the most handwriting-like texture yet: rounder and more flowing than B0 (Fig. 10). But it was not thinner than B0, and it misspelled conjuncts and long words (Table 11).

Table 11: probe_day1 (24 image_2 words; locator CER, mean). Real CER is circular: the "confident" crops were selected by the same locator. Indicative only.

method	n	width/core (median)	darkness (median)	locator CER
real (confident crops)	9	0.064	1.00	0.00 (circular)
B0	24	0.063	0.97	0.184
DP, image_2 refs	24	0.091	1.00	0.973
DP, writer_99 refs	24	0.110	1.00	0.931
DP, writer_117 refs	24	0.104	1.00	0.899
DP, fine-tuned on image_2	24	0.100	1.00	0.651

Source: `scribe_wordgen/runs/probe_day1/REPORT.md` ("Numbers")

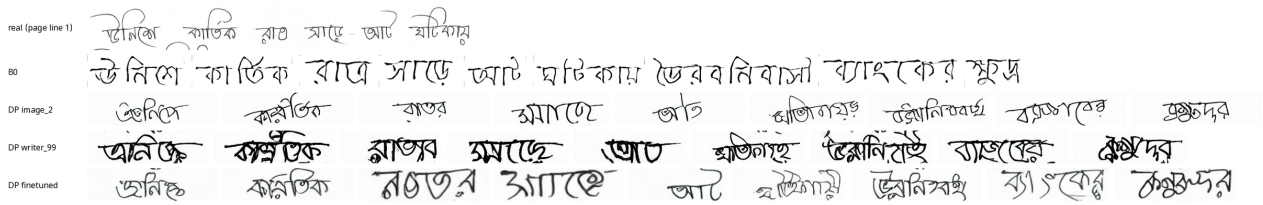


Figure 10: probe_day1: the first line of image_2, rendered by different methods (real, B0 and DiffusionPen variants).
Source: `scribe_wordgen/runs/probe_day1/line.png`

6.3 P1a: the painted content map is inert (NO-GO)

P1a trained DiffusionPen from IAM on TRAIN writers (40 crops/writer: 14,155 crops, 5,033 unique texts), 20k steps at batch 32. Arm A is the text-only control; arm B adds an 8-channel grapheme content map with a zero-initialised first-conv slice. The review first corrected the metric. Raw CER > 1 and width/core ≈ 4 were artefacts of uncapped CER on punctuation, grey-box output on grey-paper writer 117, and truncation of B0 at 256 px. Table 12 gives the corrected numbers.

Table 12: P1a, corrected (VAL, 20 writers; CER capped at 1, left-aligned, punctuation excluded).

method	CER	BN-HTR excl. 117	BW	bg-fail %	sub-px width / real
real	0.133 [0.101, 0.169]	0.049	0.237	1.2	1.00
B0	0.130 [0.105, 0.157]	0.084	0.224	0.2	≈1.3
B3	0.422 [0.384, 0.460]	0.427	0.476 (optimistic)	3.8	1.32
A	0.482 [0.440, 0.528]	0.423	0.525	8.7	1.40
B	0.457 [0.417, 0.501]	0.361	0.516	8.8	1.40

Paired B−A $= -0.024$ [-0.045, -0.002] (14/20 writers), entirely in-vocabulary (OOV: A 0.568, B 0.574).
Source: `scribe_wordgen/runs/P1a/REVIEW.md` §1

Verdict: NO-GO for the content map. When another word's map is painted in, the model still writes the CANINE word: CER against the painted word is 0.959. The map alone gives near-unconditional output (0.953 vs 0.973). The grapheme embeddings barely moved during training (row norm 0.078 vs init ≈0.057). One content patch contributes about 2% of the first-conv input signal. The review traced the other failures to grey reference paper (6.4% of BN-HTR TRAIN crops have paper < 250), a starved data set, and a CFG-driven thick pen (arm A: 1.31× at g1, 1.48× at g2/g3).

6.4 P1a′: full data, white paper, grapheme tokens and the CFG-null bug

P1a′ applied the review's fixes. It trained on all of TRAIN (98,856 crops) for 40k steps with white-paper normalisation, and redesigned content as *grapheme-token cross-attention*: learned grapheme embeddings plus column-centre and span encodings, appended to the CANINE context. The report concluded that B′ was a NO-GO (CER 0.550, worse than A′). The review found the cause.

The CFG-null bug. `eval_v2.sample_arm2` put the content map and tokens into *both* classifier-free guidance branches. Guidance therefore extrapolated only the text direction, and content was never guided. The correct null drops text, map

and tokens together, matching the independent 10% drops used in training. With the joint null (same 781 words, refs and seeds; Table 13), B′ becomes the best diffusion arm. The shuffled-content gate passes: the model writes the *painted* word (CER 0.432 against it at g3). Without any text, content alone writes the word at 0.199. The content-path ablation shows that the tokens carry identity (tokens alone 0.338), while the painted map is still inert (map alone 0.902).

Table 13: P1a′ on full VAL with the fixed sampler (eval_v2 scoring; CER capped at 1, punctuation excluded).

method	CER all	in-vocab	OOV	BN-HTR	BW	bg-fail	pen (sub-px/core)
real	0.133	0.129	0.150	0.055	0.159	1.2%	1.00
B0	0.130	0.116	0.235	0.085	0.145	0.2%	1.15
B3	0.421	0.400	0.508	0.410	0.425	3.8%	1.13
A′ g2 (as reported)	0.334	0.312	0.496	0.237	0.367	5.4%	1.36
A′ g3	0.245 [0.205, 0.287]	0.228	0.384	0.160	0.273	5.2%	1.22
B′ as reported (buggy CFG)	0.550	0.537	0.655	0.405	0.599	0.9%	–
B′ fixed g2	0.262 [0.218, 0.304]	0.239	0.385	0.153	0.298	1.3%	1.27
B′ fixed g3	**0.188** [0.150, 0.226]	0.169	0.301	0.108	0.214	1.8%	1.29
B′ content only, c4	0.199 [0.161, 0.238]	0.186	0.303	0.112	0.229	1.7%	–

Paired B′g3−A′g3 $= -0.057$ [-0.089, -0.032] (18/20 writers). A′ vs P1a A: -0.147 [-0.204, -0.103] (20/20).
Source: `scribe_wordgen/runs/P1a2/REVIEW.md` §1–2

A′ and B′ draw about 1.3× the writer's pen relative to letter size. That is thicker than B0/B3 (about 1.15), probably because P1a′ training is 84% BN-HTR, whose pens are thick. Writer 117's faint grey blotches (45–48% of its words) were not caught by the bg-fail metric, which motivated the faint-blotch score.

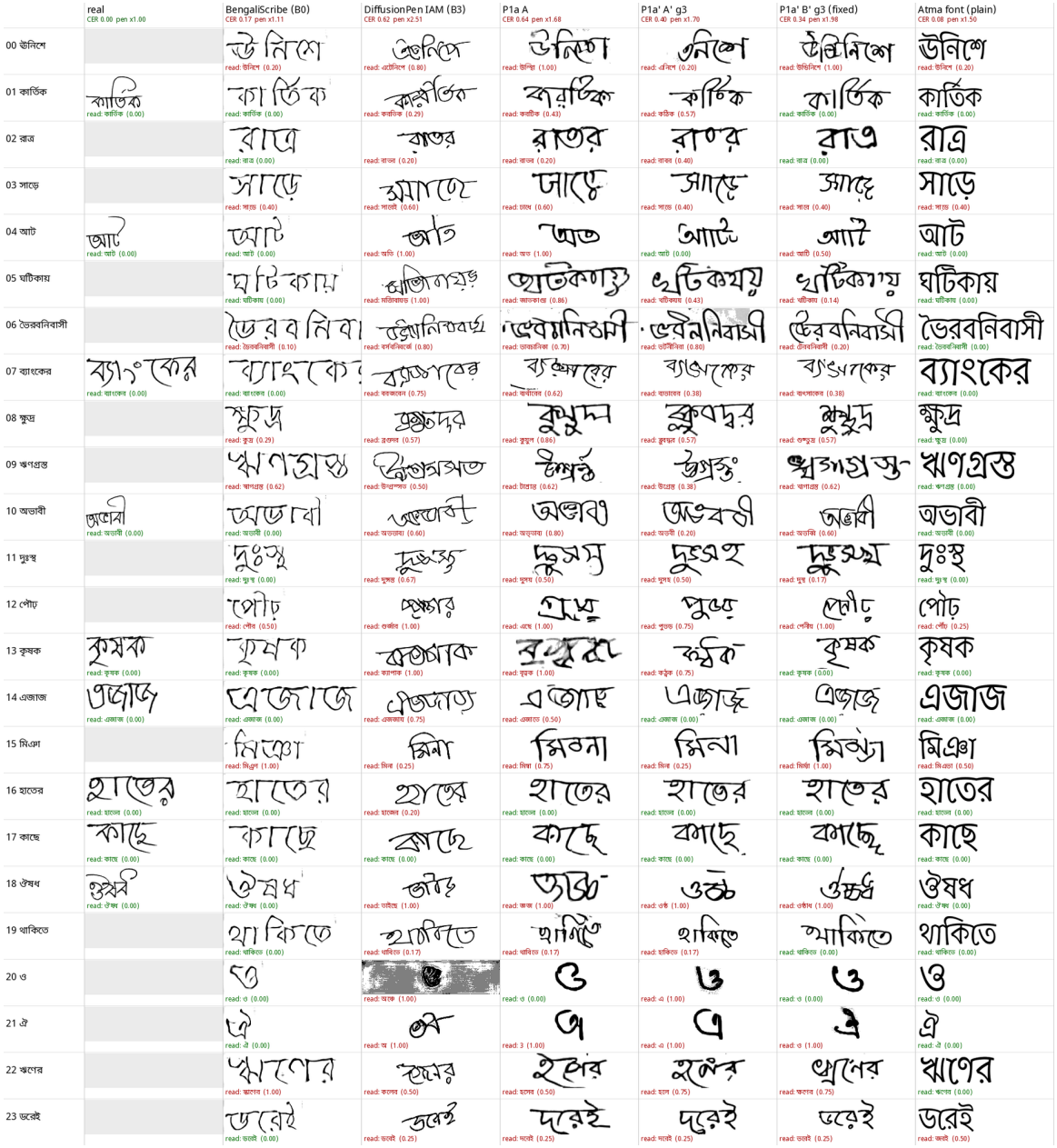


Figure 11: Side-by-side on image_2 (24 words, seed 0; the real crop exists for 9 words). Columns: real, B0, B3, P1a A, P1a′ A′ g3, B′ g3 (fixed CFG), plain Atma font. The locator's reading and CER are printed under each cell. image_2 CER over 3 seeds: B0 0.17, B3 0.62, A 0.64, A′ 0.40, B′ 0.34, Atma 0.08; pen vs writer 1.11 / 2.51 / 1.68 / 1.70 / 1.98 / 1.50.

Source: `scribe_wordgen/runs/compare_inference/image2_side_by_side.png`, `image2_scores.txt`, `README.md`

6.5 eval_v3: re-baseline and guidance sweep

eval_v3 is the corrected evaluation. It uses the fixed joint CFG null and separate text (t) and content (c) guidance scales. The decomposition is $\epsilon = \epsilon_{00} + c\,(\epsilon_{0c} - \epsilon_{00}) + t\,(\epsilon_{tc} - \epsilon_{0c})$, where the subscripts mark which conditioning (text, content) is present. eval_v3 also adds $\geq$2 seeds per word, the sub-pixel pen metric, a non-punctuation background gate, the faint-blotch score, and OOV CER on all 20 writers. Seed 0 reproduces the review exactly: B′ t3c3 gives 0.188 with 781/781 pixel-identical renders (`runs/eval_v3/validate.txt`). Raising guidance lowers CER monotonically to t5c5, which ties B0 (Table 14, Fig. 12). Pen rises slightly with guidance, and the writer-117 blotch rate falls from 34% (t2c2) to 11% (t5c5).

Table 14: eval_v3 guidance sweep for B′ (VAL, 2 seeds; locator CER).

method	CER	in-vocab	OOV (18 wr.)	BN-HTR	BW	pen [CI]	blotch 117
real	0.133 [0.100, 0.167]	0.129	0.150	0.055	0.159	1.00	8%
B0	0.130 [0.103, 0.156]	0.116	0.235	0.085	0.145	1.15 [0.98, 1.20]	0%
A′ g3	0.254 [0.214, 0.295]	0.234	0.379	0.161	0.285	1.22	31%
B′ t2c2	0.274 [0.229, 0.317]	0.255	0.393	0.143	0.318	1.26	34%
B′ t3c3	0.194 [0.156, 0.230]	0.174	0.307	0.098	0.225	1.26	21%
B′ t4c4	0.165 [0.128, 0.202]	0.151	0.254	0.077	0.194	1.27	14%
B′ t5c5	0.148 [0.115, 0.182]	0.133	0.241	0.068	0.175	1.30 [1.15, 1.45]	11%
B′ t6c6	0.143 [0.110, 0.178]	0.126	0.247	0.064	0.170	1.31	11%
B′ t0c4 (content only)	0.192	0.175	0.303	0.098	0.223	1.27	18%

Paired: B′t5c5−B0 +0.018 [-0.008, +0.044] (tie); B′t6c6−B′t5c5 −0.005 [-0.011, +0.000]; OOV B′t5c5−A′g3 −0.138 [-0.218, -0.066] (14/18 writers).

Source: `scribe_wordgen/runs/eval_v3/report_tables.txt`

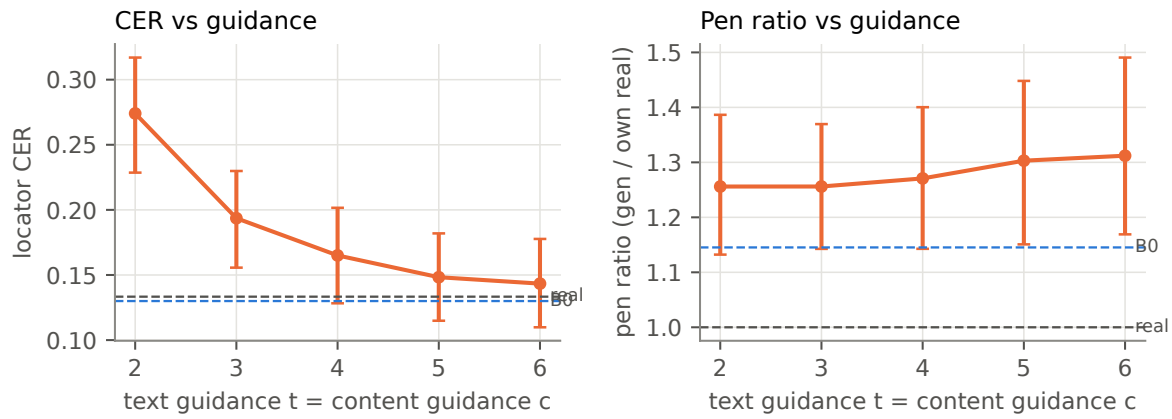


Figure 12: Guidance sweep for B′ with $t = c$. Left: locator CER with writer-bootstrap CI (dashed: real and B0). Right: pen ratio. More guidance buys legibility, and pen thickens a little.

Source: `scribe_wordgen/runs/eval_v3/metrics.json`

6.6 The style evaluator (M2 / M2s)

Before E_eval existed, style was judged with the generator's own frozen IAM encoder (real-vs-real MRR 0.41), which is not independent. E_eval and E_shape (§4) were built and checked before any style claim was made (Table 15). The first result was sobering: **every diffusion arm up to B′ sat at MRR 0.40–0.45**, far below B0 (0.596) though above fonts.

Table 15: Style evaluator validity and early arms (M2, E_eval, within-dataset galleries, 20 writers / 726 queries).

method	top-1	MRR	margin
chance	0.100	0.280	–
real	0.723 [0.651, 0.786]	0.821 [0.775, 0.859]	0.094
real via `crop_ink`	0.707	0.810	0.091
B0	0.442 [0.310, 0.591]	0.596 [0.488, 0.711]	−0.010
P1a A / P1a B	0.214 / 0.195	0.420 / 0.404	−0.133 / −0.130
A′ g3 / B′ g3	0.244 / 0.238	0.437 / 0.431	−0.114 / −0.124
NotoSans font / Atma font	0.107 / 0.092	0.289 / 0.268	−0.191 / −0.247

Paired: real−B0 +0.225 [+0.116, +0.335]; B′g3−B0 −0.165 [-0.287, -0.047]; B′g3−Noto +0.142 [+0.033, +0.247]. BN-HTR only: real 0.916, B0 0.883 (real−B0 n.s.).

Source: `scribe_wordgen/runs/style_eval/tables.md` ("eval within all", "eval within bnhtr")



Figure 13: Style-evaluator inputs after the shared preprocessing. Columns: a conditioning reference, the real target, B0, B′ g3, A′ g3, NotoSans; two variants per row.

Source: `scribe_wordgen/runs/style_eval/sheet_prep.png`

6.7 Pen calibration (post hoc, no retraining)

Generation preserves the writer's pen *rank*: Spearman over writers is about 0.9 in pixels. So a per-writer gain is well posed. `wordgen/pen/calibrate.py` re-renders strokes at page scale toward the refs' measured width. Table 16 compares the modes on B′ t5c5. All modes bring the pen into 1.0 ± 0.1. The chosen shrink mode (1.30→1.05) misses the CER gate (upper CI of the CER change $\leq +0.01$) by 0.0003. Only the per-writer mode passes both gates, but its p90 over writers is 2.20.

Table 16: Pen calibration modes on B′ t5c5 (VAL; pen median over writers, CI of median).

mode	pen ratio [CI]	p10–p90	CER after−before	gates (pen / CER)
before	1.30 [1.15, 1.45]	1.06–1.85	–	–
writer	1.09 [0.99, 1.29]	0.87–2.20	+0.0027 [-0.0030, +0.0083]	pass / pass
writer_v2	1.06 [0.92, 1.21]	0.80–1.51	+0.0049 [-0.0007, +0.0101]	pass / fail
global_v2	1.07 [1.00, 1.26]	0.85–1.52	+0.0046 [-0.0011, +0.0103]	pass / fail
word_v2	1.05 [0.92, 1.18]	0.78–1.53	+0.0053 [-0.0011, +0.0120]	pass / fail
shrink_v2 (adopted)	1.05 [0.97, 1.18]	0.83–1.37	+0.0051 [-0.0000, +0.0103]	pass / fail by 0.0003

Source: `scribe_wordgen/runs/pen_calib/report_Bp_t5c5_pencal_*.txt`

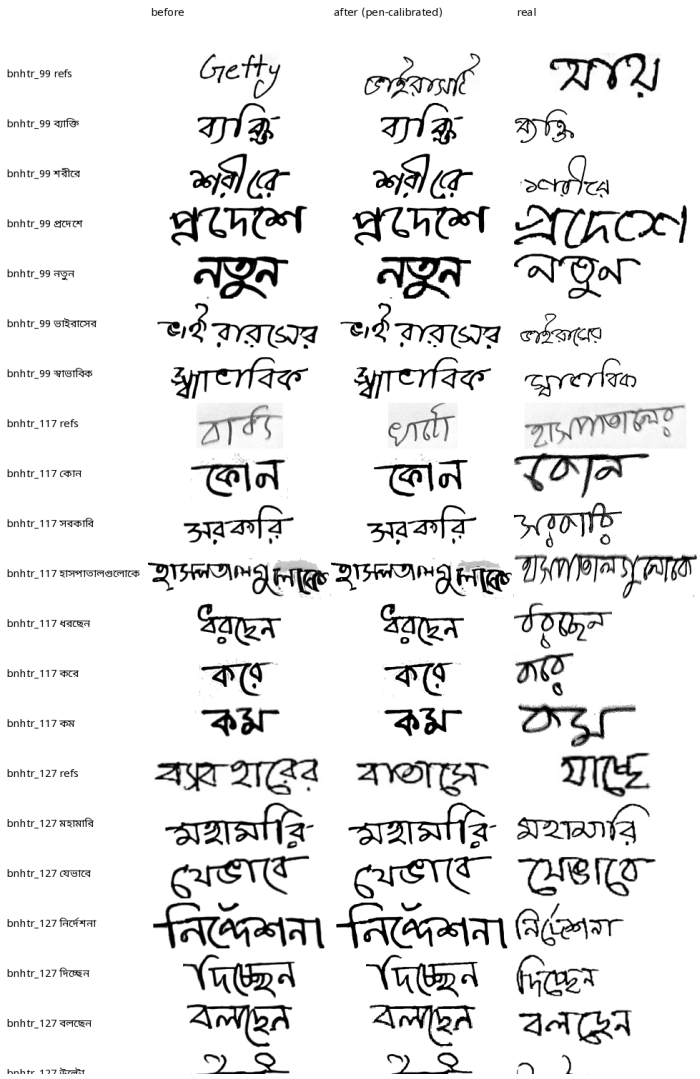


Figure 14: Pen calibration, shrink mode: B′ t5c5 before (left), after (centre) and the writer's real crop (right).
Source: `scribe_wordgen/runs/pen_calib/sheet_Bp_t5c5_pencal_shrink_v2.png` (top part)

6.8 bgnorm2: background normalisation v2

The single-scalar whitening (90th-percentile paper → 255) left writer 117's references with a grey, graded box, which the model reproduced as texture. `fit_canvas_white2` works on luminance only. It divides by a large-kernel background estimate and soft-clips the paper to exactly 255, keeping ink darkness linear. On TRAIN crops it changes ink mass by a median ratio of 0.998. It needs no retraining; the references must be passed through it (Table 17).

Table 17: bgnorm2 on B′ g3 (references only, no retraining).

method	CER	bg-fail (non-punct)	faint blotch (mean)	blotch, writer 117
real	0.133 [0.100, 0.167]	0.0%	0.4%	7.5%
B′ g3, v1 refs	0.188 [0.150, 0.226]	0.8%	3.7%	30.0%
B′ g3, bgnorm2 refs	0.183 [0.142, 0.226]	0.3%	1.8%	0.0%

Paired CER (v2−v1 refs) −0.0042 [-0.0170, +0.0079] (11/20 writers). Writer 117 alone: CER 0.080→0.116, blotch 30%→0%.
Source: `scribe_wordgen/runs/bgnorm2/bprime_noretrain.txt`, `measure.txt`

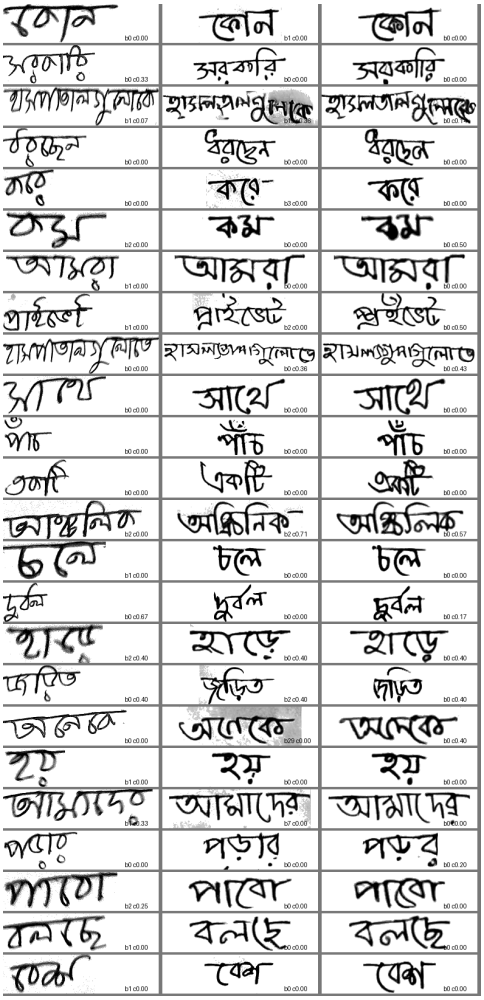


Figure 15: Writer bnhtr_117 (grey-paper scans). Columns: real crop, B′ with v1 refs, B′ with bgnorm2 refs. Each cell is labelled with blotch (b) and CER (c).

Source: `scribe_wordgen/runs/bgnorm2/sheet_gen_bnhtr_117.png` (builder `scratch/bgnorm2_gen_sheet.py`)

6.9 Arm P and Arm S

Both arms warm-start from the B′ checkpoint and train on 100,868 TRAIN rows encoded with bgnorm2. Training uses batch 32, lr 10^{-4} (10× for new modules), 300 warm-up steps and a final lr fraction of 0.1. Dropouts are text, content and style 0.1 and pen 0.05, and the CFG null drops all of them together. Sampling is rebalanced to punctuation 1.5%, BW 38% and BN-HTR 60.5%, against a natural mix of 2.0 / 15.4 / 82.6%. Writers within a dataset are weighted by clip(median pen / writer pen, 0.5, 2.0), which up-weights thin pens. The two arms differ as follows (`runs/P1a3/{P, S}/config.json`):

- **Pen knob** (both). The input is the refs' median $\log(w_{\text{sub}}/c)$, z-scored with TRAIN statistics ($\mu = -1.835$, $\sigma = 0.342$) and embedded sinusoidally. A zero-initialised MLP adds it to the timestep embedding, so the knob is a no-op at initialisation.
- **Null-style fix** (both). In B′, dropped style fell back to the TRAIN writer-ID label embedding. P and S instead use a learned null style vector.
- **Reference tokens** (S only). A Bengali writer encoder (ResNet-18, SupCon on TRAIN) turns each ref into 16 spatial tokens. With slot embeddings, 5 refs give 80 tokens. They enter every transformer block through decoupled attention, out $=$ attn$(q,$ text+content$)$ $+$ keep $\cdot\, W_{\text{ref}}$ attn$(q, K_{\text{ref}}, V_{\text{ref}})$. W_{ref} starts at zero, which keeps the no-op property. Appending the tokens to the shared context would not: 80 extra keys change the softmax normalisation even with zero weights.

P trained for 16k steps (4,012 s) and S for 24k steps (2,109 s; final EMA loss 0.0581).

Two diverged S attempts and the gradient bug. The first two S runs diverged. Run 1's rolling loss jumped from ≈0.065 to above 0.14 within ∼800 steps. Run 2 drifted up from ≈0.064 to 0.108 by step 1,050 and was stopped (Fig. 16). The cause was a gradient bug. DiffusionPen's `CheckpointFunction` re-runs each transformer block during backward. The side-channel attributes (`n_ref` and the keep mask) were reset after the forward pass, so the recomputation was a different function: full-context attention without the reference branch. S therefore trained on wrong gradients. The fix keeps these attributes set; a test checks that S's gradient matches a run without checkpointing (`wordgen/models/diffusion3.py`, `tests/test_p1a3.py`). The final S config also adds gradient clipping 1.0 and a 2× reference-token lr multiplier, which the diverged configs lack.

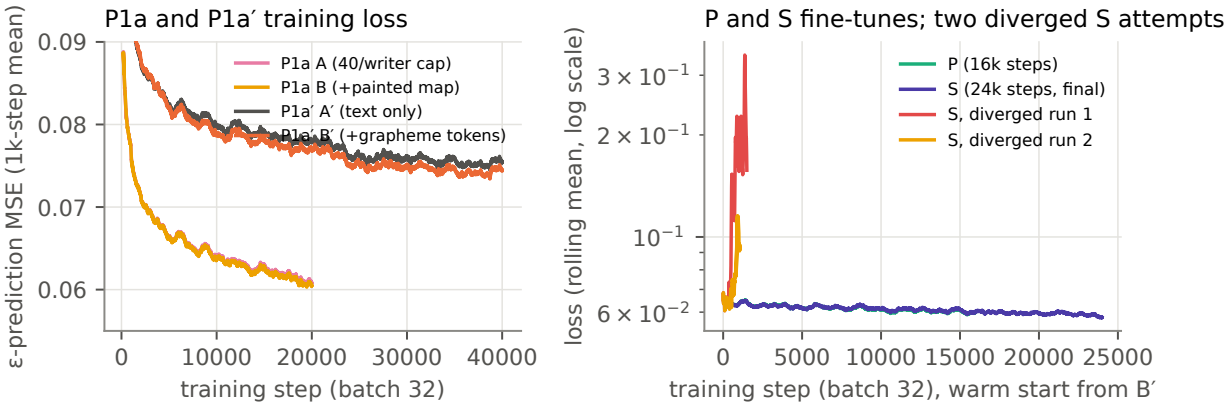


Figure 16: Training curves. Left: P1a (capped data) and P1a′ (full data). The full-data loss level is higher because the data are harder and more varied; the curves are not comparable across data sets. Right: P and S fine-tunes from B′, and the two diverged S runs (log scale).

Source: `scribe_wordgen/runs/P1a{,2}/{A,B}/metrics.csv`, `runs/P1a3/{P,S,S_diverged_v1,S_diverged_v2}/metrics.csv`

Table 18: P and S on VAL (locator CER, 2 seeds; background and pen). _cal = shrink-mode pen calibration.

method	CER	1–2 graph	>2 graph	OOV	bg-fail non-punct	blotch 117	pen raw [CI]
real	0.133 [0.100, 0.167]	0.130	0.135	0.150	0.0%	8%	1.00
B0	0.130 [0.103, 0.156]	0.115	0.143	0.235	0.0%	0%	1.15 [0.98, 1.20]
B′v2 t5c5	0.151 [0.117, 0.186]	0.146	0.157	0.246	0.6%	1%	1.29 [1.18, 1.46]
P t3c3	0.157 [0.118, 0.198]	0.185	0.128	0.226	3.0%	0%	1.13 [1.02, 1.28]
P t5c5	0.132 [0.097, 0.171]	0.163	0.104	0.194	2.7%	0%	1.15 [1.01, 1.30]
S t3c3	0.153 [0.115, 0.189]	0.192	0.118	0.216	3.6%	0%	1.12 [1.05, 1.21]
S t5c5	0.137 [0.101, 0.174]	0.173	0.104	0.202	3.0%	0%	1.13 [1.07, 1.21]
P t5c5_cal	0.133	0.165	0.105	0.195	2.6%	0%	1.06 [0.97, 1.15]
S t5c5_cal	0.138	0.175	0.103	0.209	3.1%	0%	1.06 [0.98, 1.12]

Paired CER: P−B0 +0.002 [-0.028, +0.037] (10/20); S−B0 +0.007 [-0.023, +0.040] (10/20); S−P +0.006 [-0.004, +0.015]; P−B′v2 −0.019 [-0.034, -0.004] (16/20). OOV S−B0 −0.033 [-0.088, +0.016]. **The 1–2-grapheme column and bg-fail for P/S carry the `img_to_ink` short-word artefact** (§6.10).

Source: `scribe_wordgen/runs/P1a3/eval/report_tables.txt`

Table 19: Style for P and S: M2 (E_eval) and M2s (E_shape) MRR, VAL, hierarchical bootstrap.

method	M2 MRR (all)	M2 BN-HTR / BW	M2s MRR (all)	M2s BN-HTR / BW
real	0.821 [0.775, 0.859]	0.916 / 0.789	0.768 [0.724, 0.812]	0.927 / 0.715
B0	0.596 [0.488, 0.711]	0.883 / 0.500	0.533 [0.453, 0.617]	0.841 / 0.430
B′v2 t5c5	0.418 [0.355, 0.483]	0.566 / 0.369	0.399 [0.351, 0.449]	0.609 / 0.330
P t5c5	0.443 [0.379, 0.505]	0.556 / 0.405	0.416 [0.367, 0.469]	0.596 / 0.357
S t5c5	**0.579** [0.521, 0.635]	0.745 / 0.523	**0.517** [0.469, 0.564]	0.726 / 0.448
S t5c5_cal	0.562 [0.511, 0.615]	0.716 / 0.511	0.518 [0.471, 0.566]	0.729 / 0.447

Paired M2: S−B0 −0.017 [-0.116, +0.088] ($p = 0.72$, 9/20); S−P +0.136 [+0.085, +0.187] (17/20); P−B0 −0.153 [-0.261, -0.048]. M2s: S−P +0.101 [+0.057, +0.140]; S−B0 −0.015 [-0.091, +0.061]. Chance: all 0.28; BN-HTR 0.457; BW 0.221.

Source: `scribe_wordgen/runs/P1a3/eval/report_tables.txt`, `runs/P1a3/eval/metrics.json`

All P/S variants pass the planned gates from the P1a′ review: CER ≤ B0+0.05 = 0.180, calibrated pen ≤1.15, and writer-117 blotch <10%.

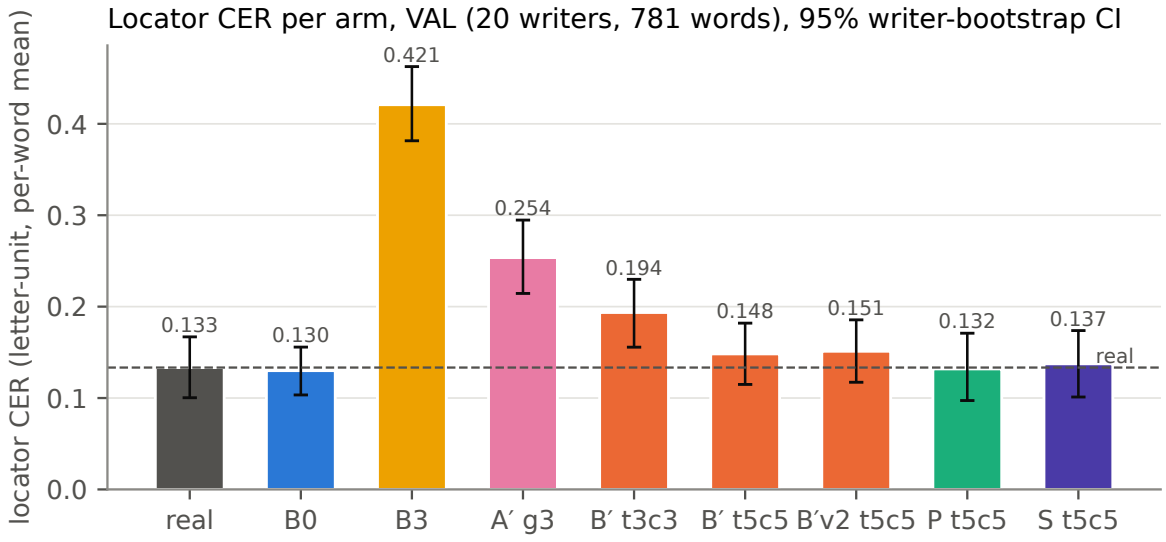


Figure 17: Locator CER per arm (letter units; capped per-word mean). From B′ t5c5 on, every diffusion arm ties B0 on this metric.

Source: `runs/eval_v3/metrics.json`, `runs/P1a3/eval/metrics.json`

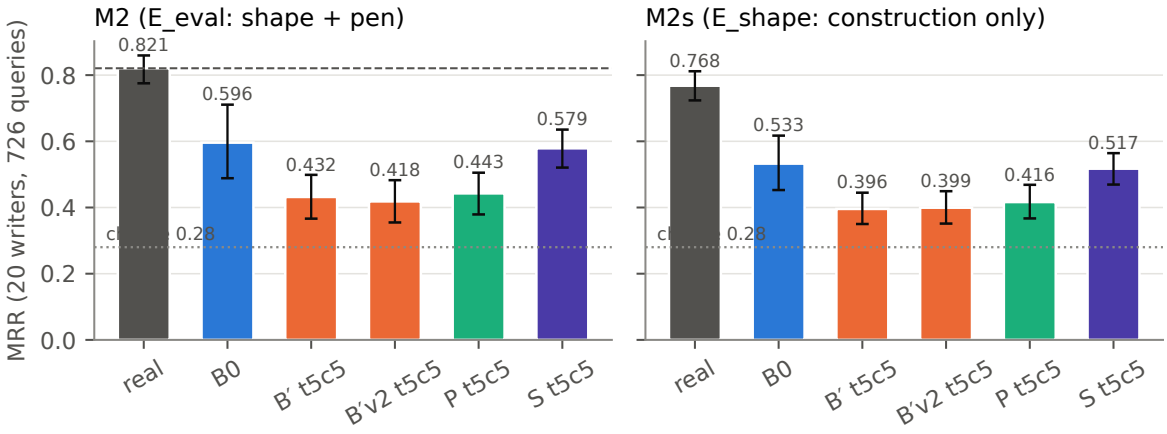


Figure 18: Style MRR per arm, with the chance line (dotted) and the real-crop level (dashed, left). The reference tokens of S lift style from the 0.42–0.44 plateau of B′/P to B0's level.

Source: `runs/P1a3/eval/metrics.json`

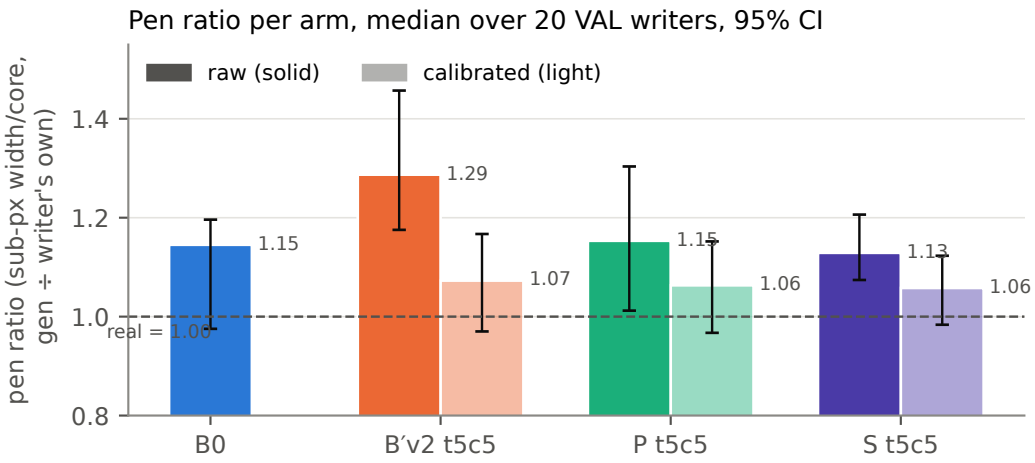


Figure 19: Pen ratio, raw and calibrated (median over 20 writers, 95% CI of the median).
Source: `runs/P1a3/eval/metrics.json`

Pen-knob sweep. The knob was multiplied by 0.6–1.6 (g5, seed 0, 8 words × 20 writers; Fig. 20). In P the within-writer slope $d\log(\text{generated})/d\log(\text{knob})$ is 0.11 [0.08, 0.14]: the knob works but is weak. In S it is 0.03 [0.00, 0.05], **nearly inert**. The reference tokens apparently carry the pen and override the scalar. Across writers without a sweep, generated width tracks the knob with slope 0.15 in both arms (Spearman 0.31 P, 0.15 S).

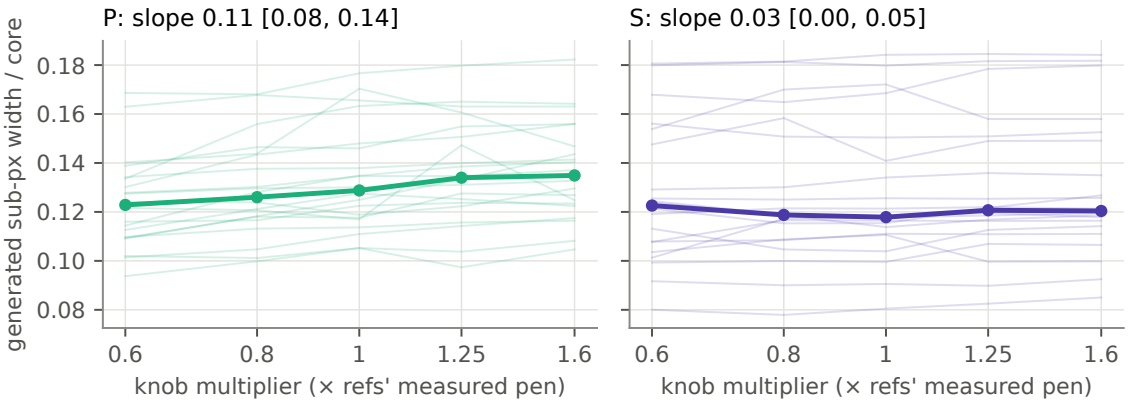


Figure 20: Pen-knob sweep. Thin lines: per-writer medians; bold: median over writers. Titles give the within-writer slope with its 95% CI over writers.
Source: `runs/P1a3/eval/sweep.csv`, `runs/P1a3/eval/metrics.json` (`knob_sweep`)

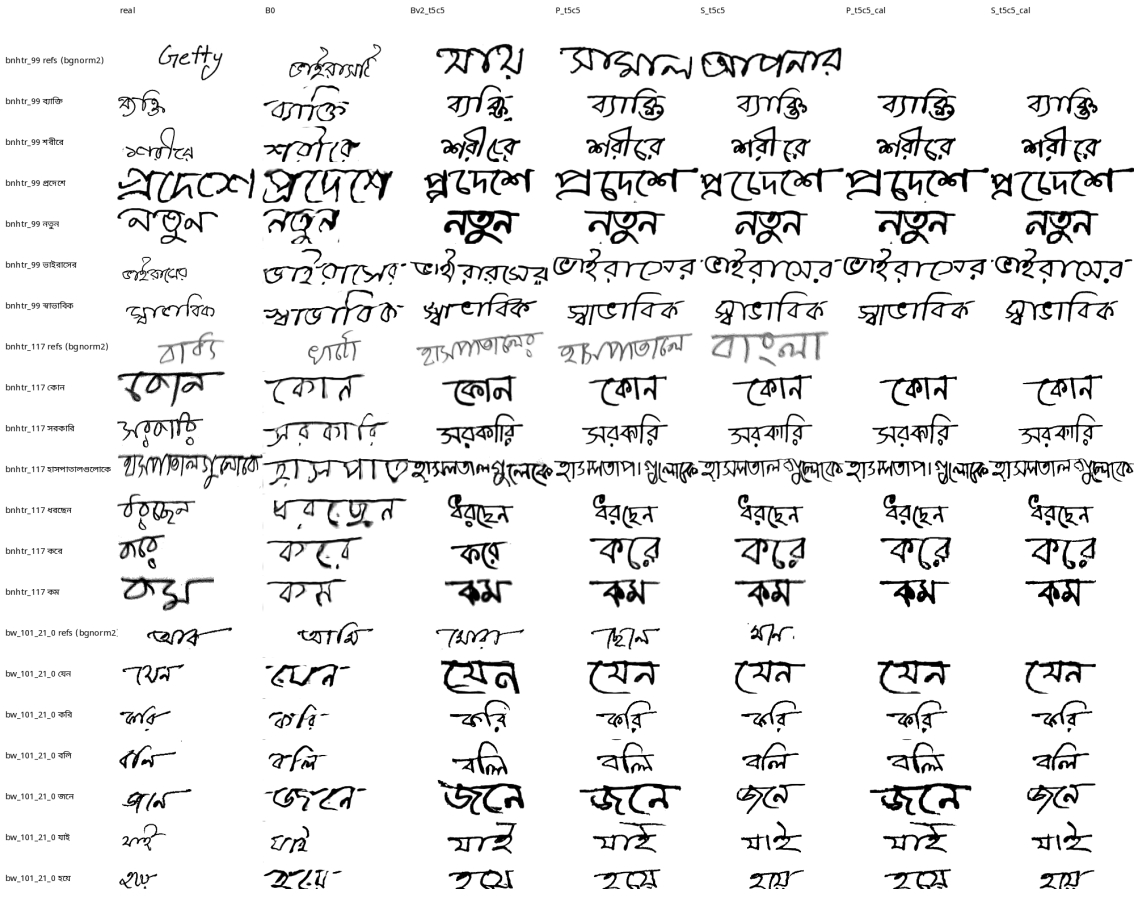


Figure 21: VAL words (top of the P1a3 sheet; seed 0). Columns: real, B0, B′v2 t5c5, P t5c5, S t5c5, P calibrated, S calibrated. “refs (bgnorm2)” rows show the writer’s conditioning crops.

Source: `scribe_wordgen/runs/P1a3/eval/sheet_val.png`

6.10 Page sheets and the smudge fix

Full pages were rendered with the learned layout (`scribe_layout_v2/runs/default_unfitw_dc/model.pt`) for B0, B′, P and S. The writers were image_2 and three VAL writers (`runs/page_sheets/`). Two problems are visible (Fig. 22). Generated text on BanglaWriting pages renders smaller than the real page, and P draws dash words (“–”) as thick bars.

The “smudges” were not the model. Isolated glyphs such as ০, ও and ২ came out as dark smudges. The cause is in `img_to_ink`. Its 5th-percentile ink floor assumes ink covers $\geq$5% of the canvas. A lone glyph covers 2–3%, so the floor lands on paper texture and a ripple of about 12 grey levels is stretched to black. The fix `img_to_ink(..., min_span=100)` keeps the floor at least 100 grey levels below paper. It brings the lone-glyph ink fraction from 39–56% down to 2–3% and leaves normal words unchanged. The fix is opt-in. **The eval caches still use the legacy conversion**, so short-word CER and blotch for the diffusion arms P, B′ and S carry the artefact until all arms are rescored together. The old inks are kept in `runs/page_sheets/cache_v1_smudged/`.

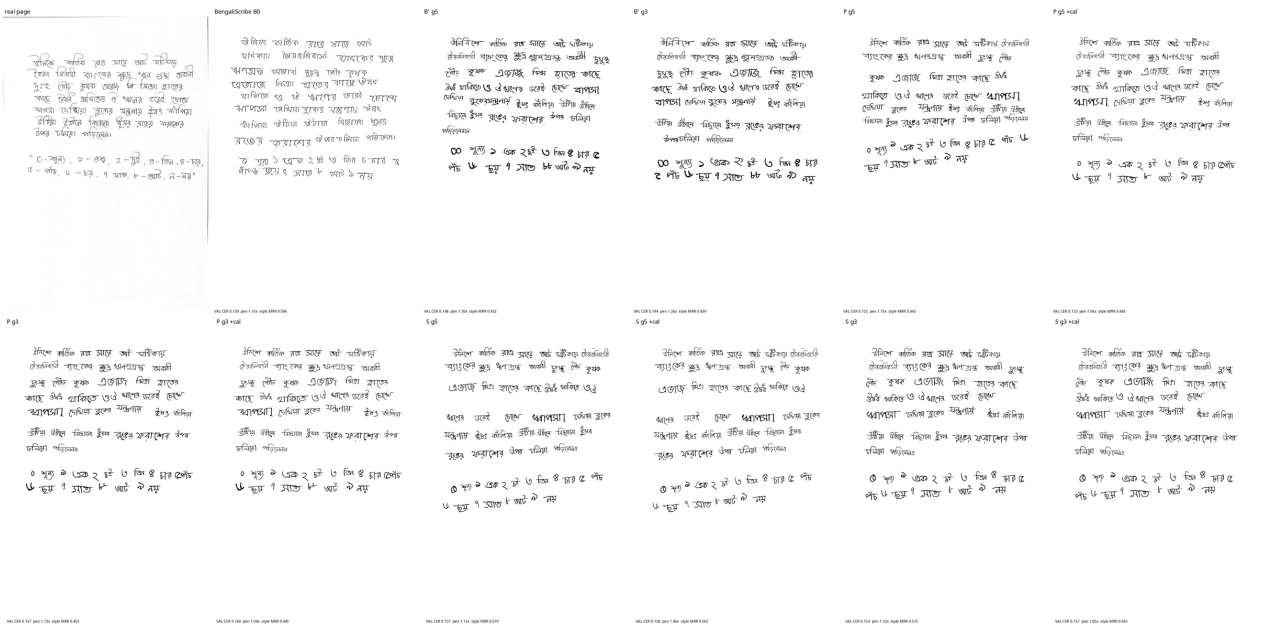


Figure 22: Page sheet for image_2, split into two rows of six panels. Row 1: real page, B0, B′ g5, B′ g3, P g5, P g5 calibrated. Row 2: P g3, P g3 calibrated, S g5, S g5 calibrated, S g3, S g3 calibrated.

Source: `scribe_wordgen/runs/page_sheets/sheets/image_2_sheet_l0.png`

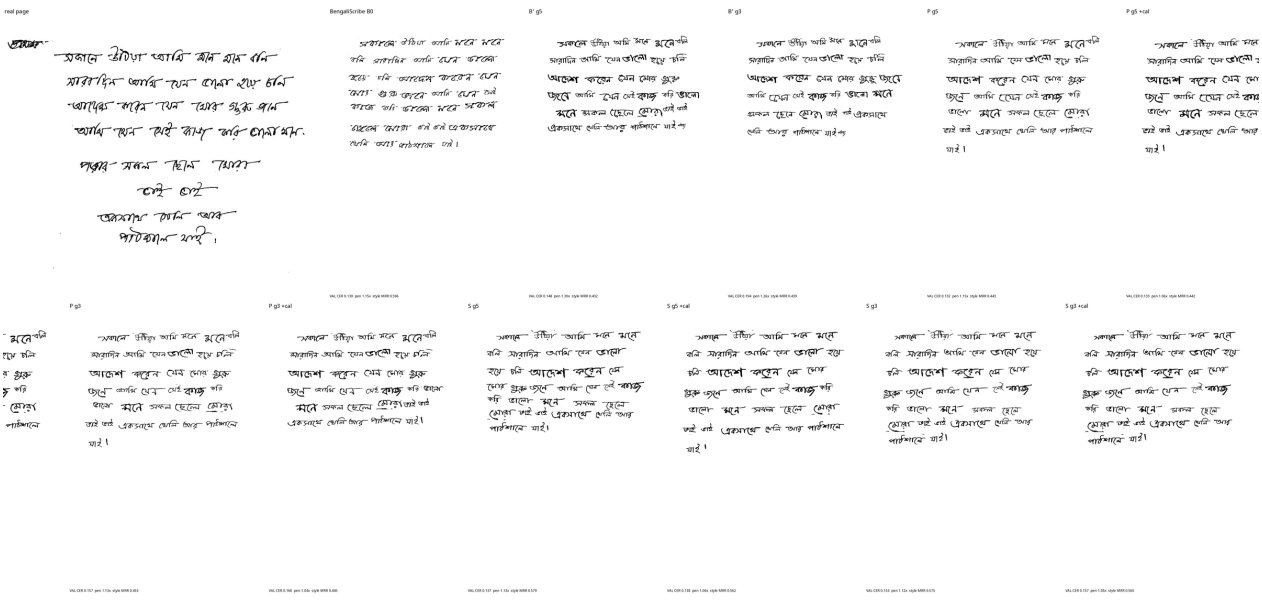


Figure 23: Page sheet for BanglaWriting VAL writer bw_101_21_0 (same panel order). The generated text is smaller than the real writer's.

Source: `scribe_wordgen/runs/page_sheets/sheets/bw_101_21_0_sheet_l0.png`

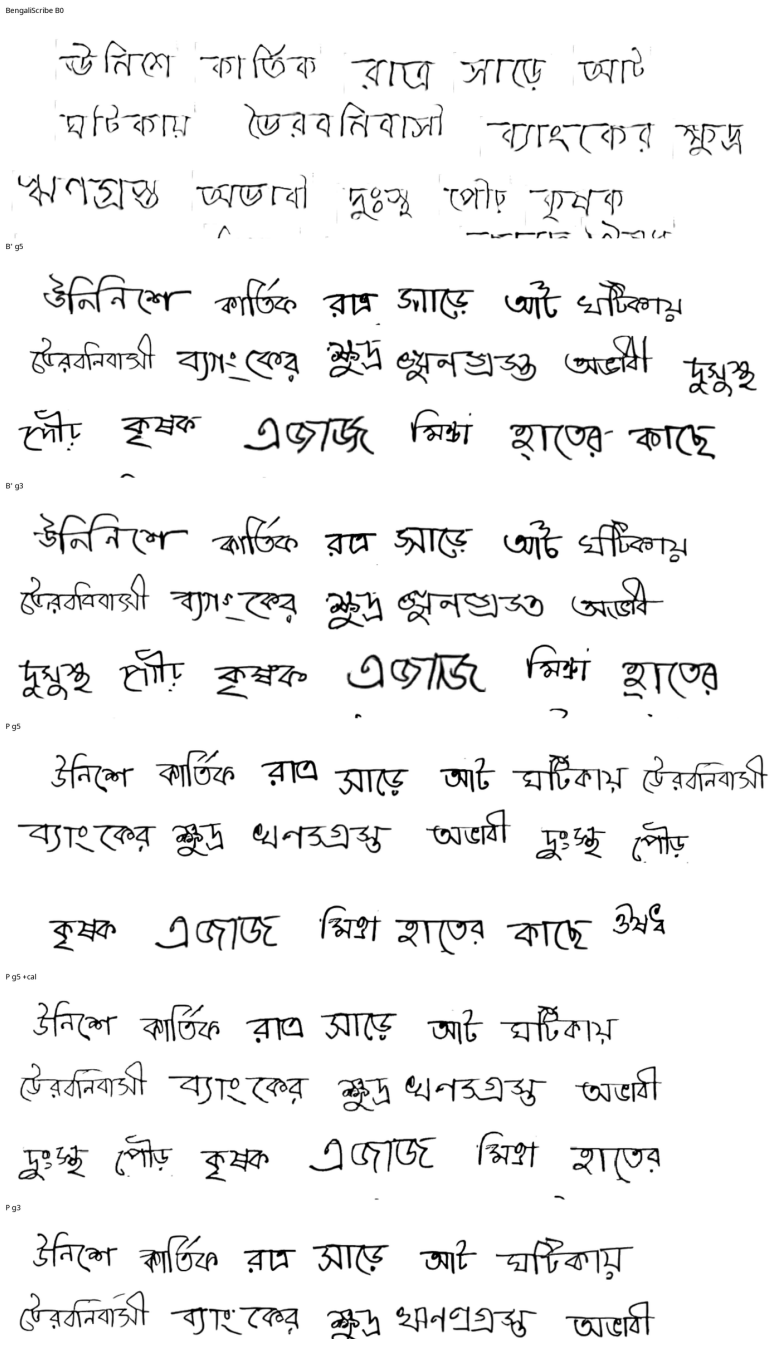


Figure 24: Zoom on the first lines of the image_2 page: B0, then B′ g5, B′ g3, P g5 and P g5 calibrated.
Source: `scribe_wordgen/runs/page_sheets/sheets/image_2_zoom_l0.png` (top part)

7 GlyphScribe fonts and the R_eval recogniser

7.1 Fonts

The first font search scored 11 Google OFL Bengali fonts against image_2 (12 hand-verified words) by skeleton chamfer. Atma, Hind Siliguri and Noto Serif Bengali came out closest, with overlapping CIs; NotoSans ranked 11th of 12 (`runs/font_search/REPORT.md`). Then 37 handwriting fonts, 1 printed font and 4 paper backgrounds were copied from the public GlyphScribe repository. Its licence covers the code only and the font licences are unknown, so these fonts are for internal research only. All 38 are Unicode Bengali with correct GSUB shaping. On shape, Atma variants rank first (chamfer 0.162–0.164) and NotoSans ranks 28th of 39. The top ∼10 fonts overlap within their CIs (Fig. 25). Thin fonts match the pen far better: the writer's width/core is 0.075, against 0.094 for Junudullah Zami and Tagore, 0.068 for Qayyum Thin and 0.11–0.23 for the Google fonts. No font draws the writer's broken matra.

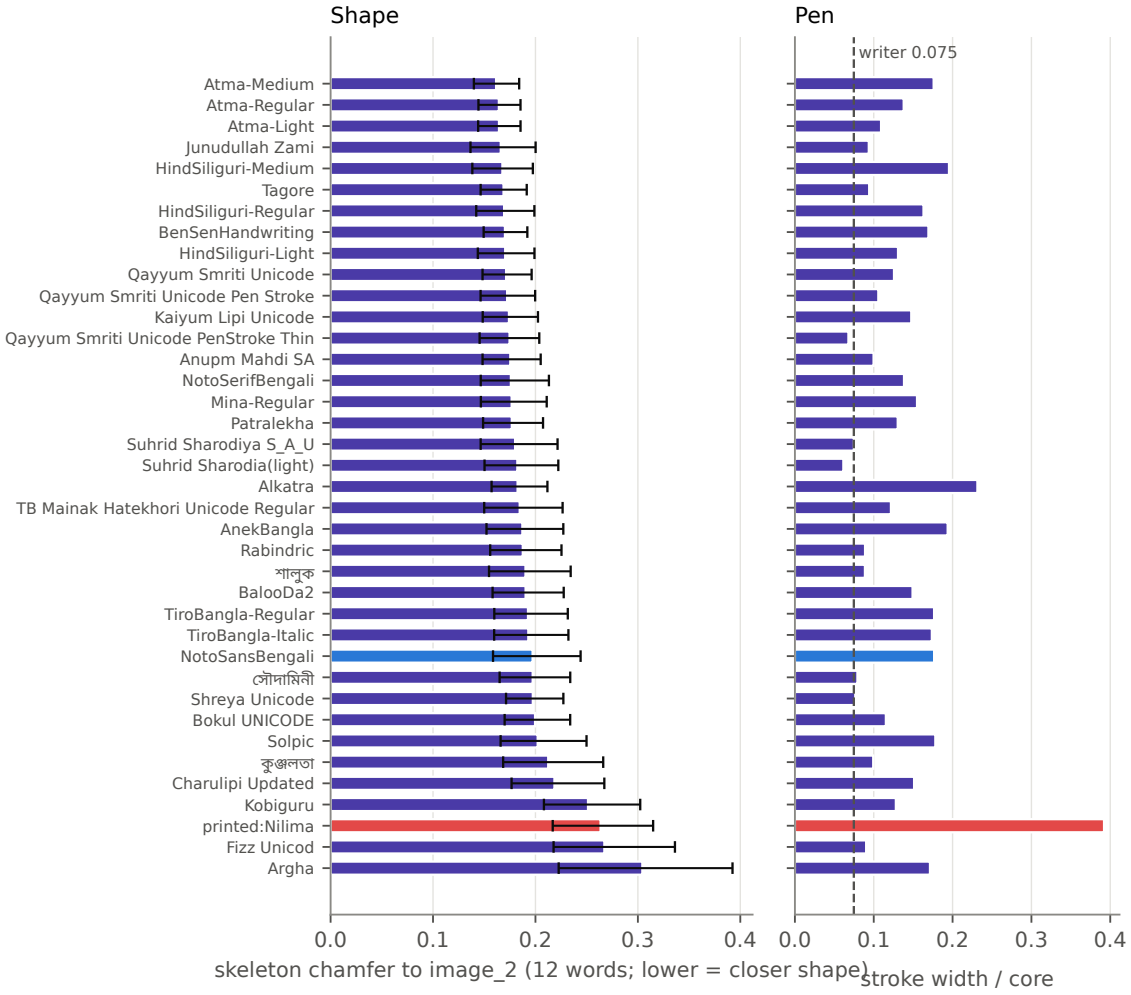


Figure 25: All scored fonts against image_2: skeleton chamfer with 95% CI over 12 words (left), stroke width/core (right). Blue: NotoSans (BS's font). Red: the printed font.

Source: `scribe_wordgen/runs/font_search/glyphscribe/scores.csv`



Figure 26: image_2's real words (left block) against the closest fonts, each shown raw and pen-matched.

Source: `scribe_wordgen/runs/font_search/glyphscribe/sheet.png`

7.2 R_eval: an independent recogniser

R_eval answers two questions. Does pretraining on synthetic renders from the 48 fonts (37 GlyphScribe + 11 Google) help on real handwriting? And how legible are the generator arms to a recogniser the generators never saw? The

model is a **CRNN-CTC** with 4.94 M parameters. The input is a 64×256 left-aligned ink matte. A conv/ResNet stack reduces it to 1×64, followed by a 2-layer BiLSTM (256 hidden) and 219 classes including blank and unk; decoding is greedy CTC (`wordgen/recog/model.py`). Synthetic words come from a 203,371-word vocabulary (`runs/R_eval/vocab_synth.tsv`). They are rendered with HarfBuzz shaping, decorative fonts are down-weighted to $0.3\times$, and the renders are augmented for thickness, shear, rotation, baseline wave, blur, noise, paper texture and JPEG. The real data are the 100,868 TRAIN crops. Before launch, a review fixed three blockers: wrong word pairing in rescoring, double `crop_ink` on cached mattes, and an augmentation roll that cut word edges. The whole chain took 1 h 50 min (`runs/R_eval/full_chain.log`).

Table 20: R_eval arms and VAL CER (codepoint, corpus; both datasets, 4,299 words; writer-bootstrap CI).

model	init	data, steps	VAL CER	BN-HTR	BW
R_font0	–	synthetic only, 60k	0.4563 [0.3265, 0.5338]	0.4827	0.3538
R_real	–	real TRAIN, 30k	0.0370 [0.0281, 0.0511]	0.0308	0.0614
R_font2real	R_font0	real TRAIN, 30k	**0.0329** [0.0245, 0.0443]	0.0283	0.0507
R_mix	R_font0	75% real + 25% synth, 30k	0.0370 [0.0267, 0.0486]	0.0333	0.0514
R_real_10	–	10% of TRAIN writers, 6k	0.1781 [0.1404, 0.2082]	0.1731	0.1976
R_font2real_10	R_font0	10% of TRAIN writers, 6k	0.1147 [0.0797, 0.1366]	0.1162	0.1090

Batch 128, lr 3×10^{-4}, warm-up 500. R_font2real is selected as R_eval. For reference, the locator's letter-unit real-crop CER is 0.056 (BN-HTR VAL) and 0.236 (BW VAL); it is not comparable.

Source: `scribe_wordgen/runs/R_eval/RESULTS.md`, `selection.json`, `R_*/train.log`

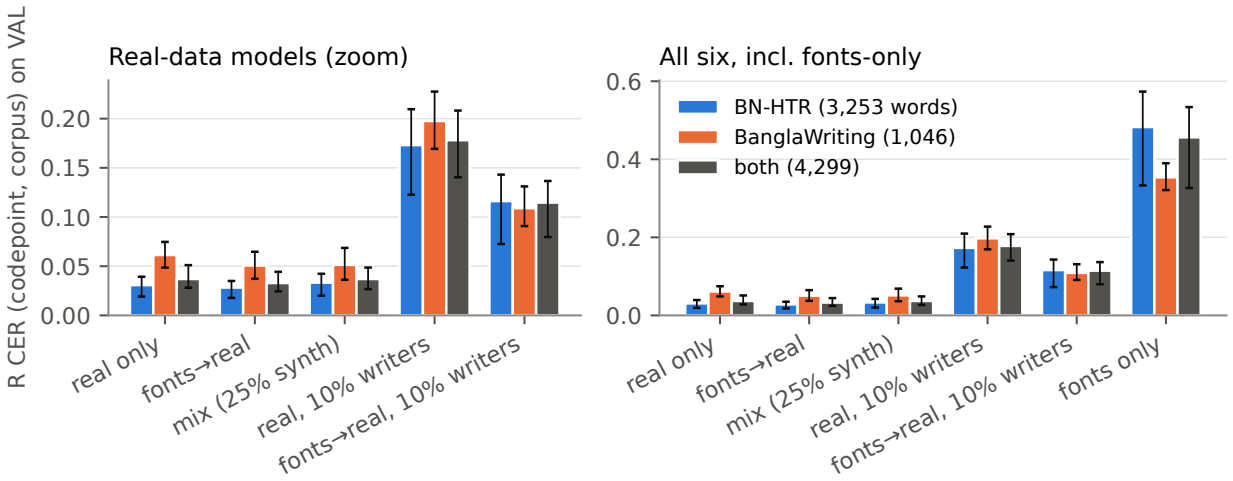


Figure 27: VAL CER per recogniser, by dataset (95% writer-bootstrap CI).

Source: `runs/R_eval/RESULTS.md`

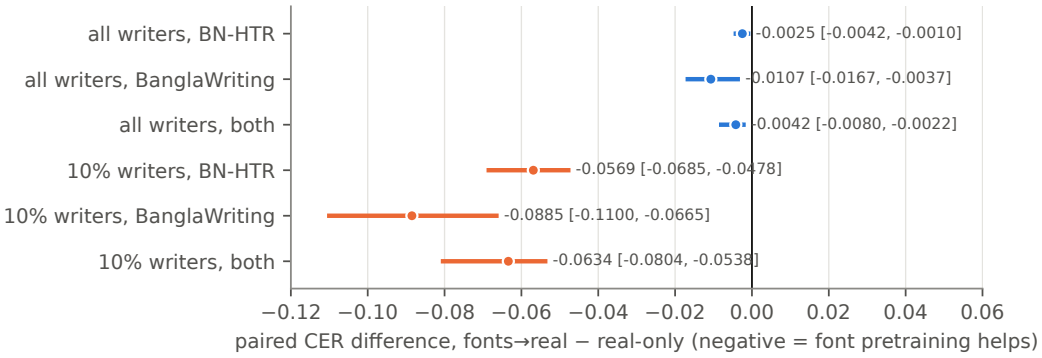


Figure 28: Font-pretraining gain, as paired differences on the same VAL words. It is significant everywhere: small with all writers, large with 10% of the writers.

Source: `runs/R_eval/RESULTS.md` (paired font-pretraining gain table)

Findings. Fonts alone do not transfer (0.456). As a prior they help: -0.0042 [-0.0080, -0.0022] overall and -0.0107 [-0.0167, -0.0037] on BW with all writers. With 10% of the writers the gain is -0.0634 [-0.0804, -0.0538], from 0.178 to 0.115. Mixing synthetic data into fine-tuning (R_mix 0.0370) does not beat plain fine-tuning. The remaining errors are mostly lone punctuation (, । ') and Latin/Bengali digit confusion (Fig. 31).

OOV and TEST. **TEST was scored once**, for R_real and the selected model: 7,665 words. Overall CER is 0.0429 [0.0293, 0.0606] for R_real and 0.0342 [0.0228, 0.0492] for R_font2real. On "clean" OOV words (not in TRAIN and not in the synthetic vocabulary, 237 words) the two give 0.0956 and 0.0854, with overlapping CIs (Fig. 29). These are two separate estimates; no paired TEST difference was computed.

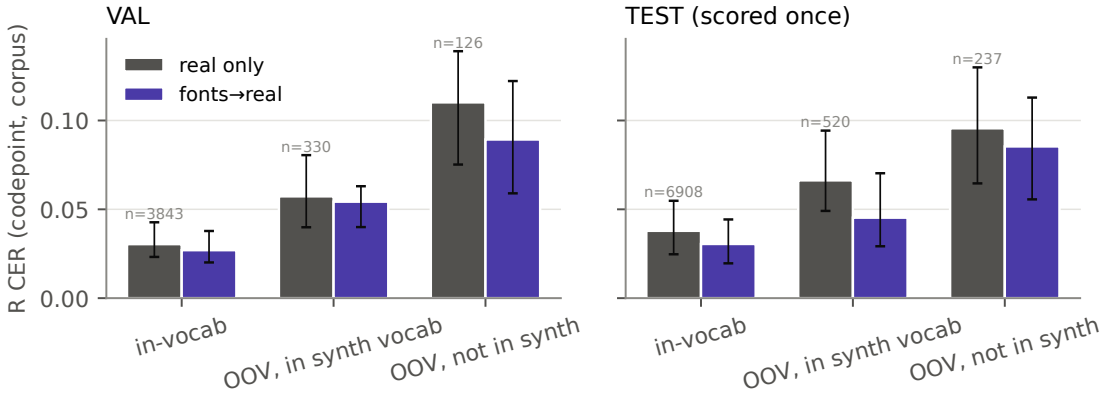


Figure 29: In-vocabulary vs OOV CER (overall). "In synth" means the word is in the synthetic vocabulary. TEST was scored once.
Source: `runs/R_eval/RESULTS.md`

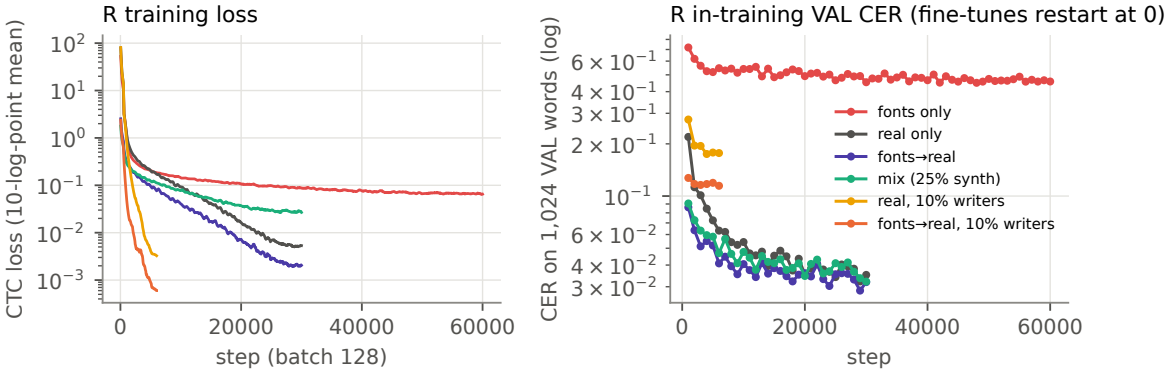
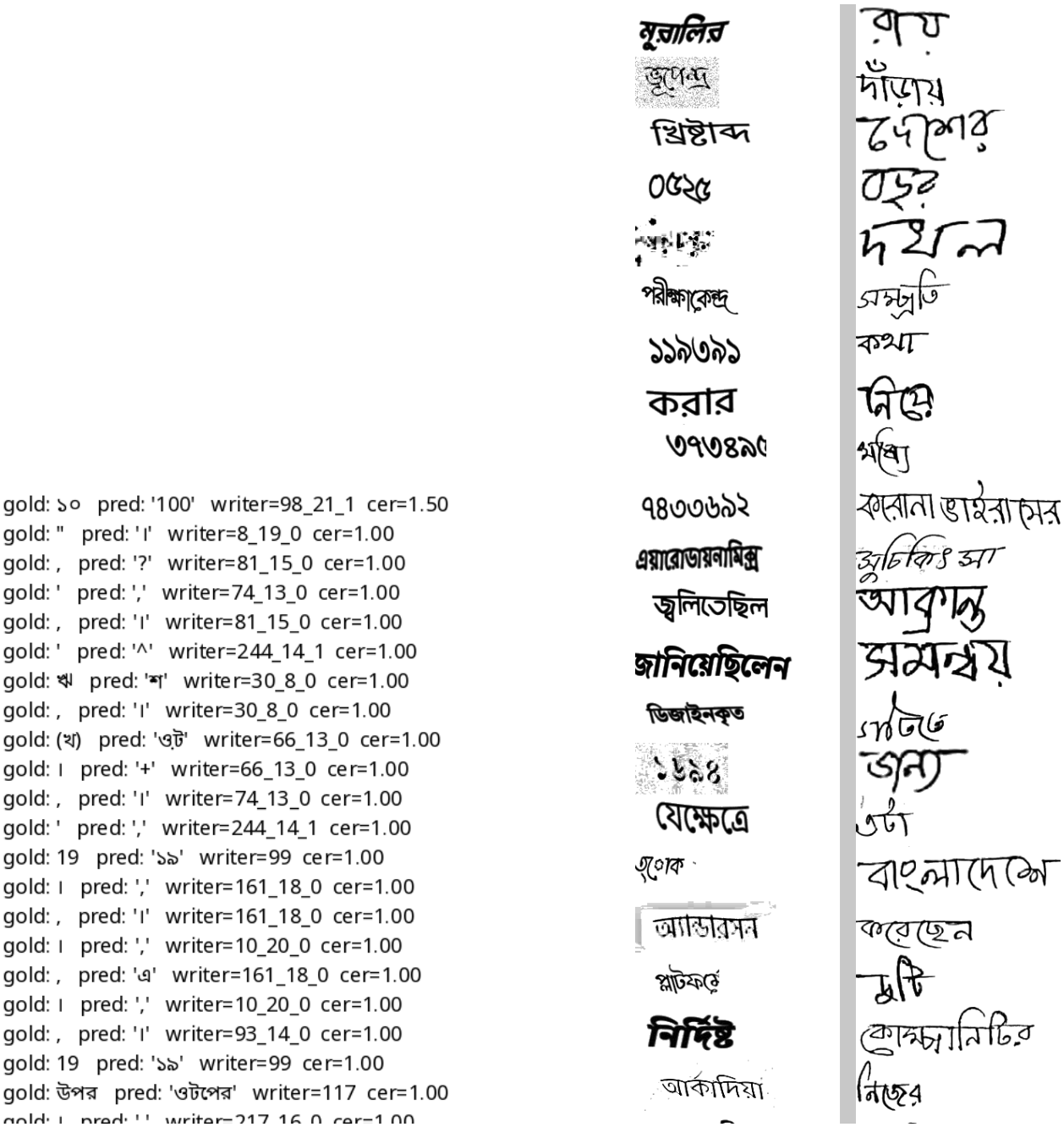


Figure 30: R learning curves. Left: CTC training loss. Right: CER on a 1,024-word VAL sample logged every 1,000 steps. This in-training estimate is not the full-VAL number of Table 20.
Source: `runs/R_eval/R_*/train.log`



(a) R_font2real, VAL: top error rows.

(b) Synthetic training renders (top of sheet).

Figure 31: R_eval errors and synthetic data. Most residual errors are single punctuation marks and digit-script confusions (19 read as ১৯).

Source: `runs/R_eval/errors_R_font2real_val.png`, `runs/R_eval/synth_sheet.png`

7.3 Rescoring the generator arms with R_eval

The cached eval_v3/P1a3 renders were rescored. They are ink mattes, so they are loaded as $1 - \text{png}/255$ with no re-stretch. Rows are merged on (dataset, writer, idx) against B0 and real, and on (dataset, writer, idx, seed_k) between diffusion arms. B0 words wider than 256 px (about 10%) are squashed to fit, whereas the locator truncated them.

Table 21: Generator arms scored by R_eval (R_font2real), VAL, codepoint corpus CER.

method	n (all)	CER all words	n ($\geq$3cp)	CER $\geq$3 codepoints
real	781	0.0438 [0.0310, 0.0586]	666	0.0424 [0.0294, 0.0568]
B0	781	0.0302 [0.0248, 0.0358]	666	0.0226 [0.0162, 0.0287]
B′ t5c5 (v1 refs)	1562	0.1395 [0.1117, 0.1713]	1332	0.1021 [0.0810, 0.1264]
B′v2 t5c5*	1562	0.1412 [0.1125, 0.1749]	1332	0.1024 [0.0823, 0.1272]
P t5c5	1562	0.0905 [0.0664, 0.1179]	1332	0.0577 [0.0402, 0.0786]
S t3c3	1562	0.1084 [0.0804, 0.1408]	1332	0.0678 [0.0501, 0.0888]
S t5c5	1562	0.0889 [0.0652, 0.1153]	1332	0.0552 [0.0381, 0.0753]
paired differences (all words; $\geq$3cp)				
S−B0		+0.0587 [+0.0361, +0.0833]		+0.0326 [+0.0145, +0.0523]
S−P		−0.0016 [-0.0111, +0.0076]		−0.0025 [-0.0099, +0.0050]
P−B0		+0.0603 [+0.0375, +0.0860]		+0.0351 [+0.0174, +0.0567]
B0−real		−0.0136 [-0.0268, -0.0021]		−0.0198 [-0.0347, -0.0067]

*B′v2 is not in `RESULTS.md`; its row is from the full rescoring table in `runs/R_eval/full_chain.log`.
Source: `scribe_wordgen/runs/R_eval/RESULTS.md` ("scored with R_eval"), `runs/R_eval/rescore.csv`

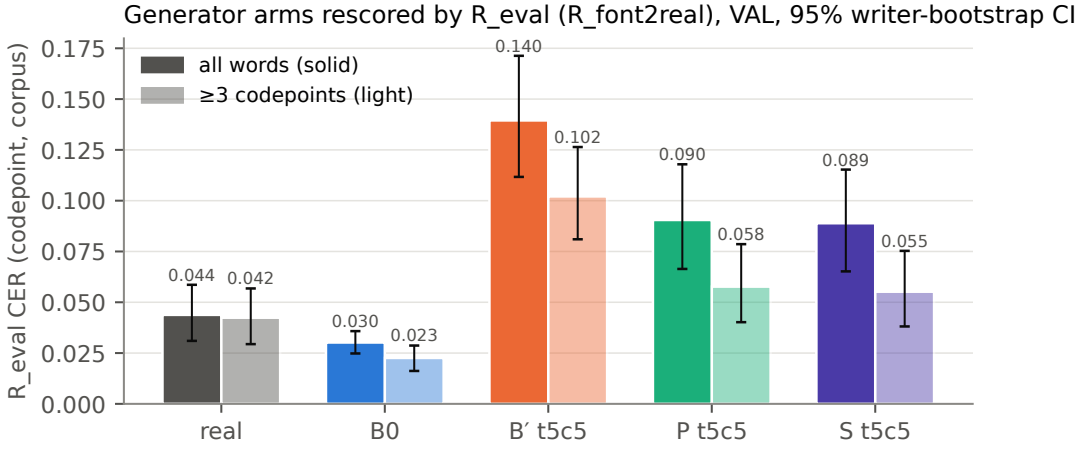


Figure 32: R_eval CER per arm. Solid: all words. Light: words of $\geq$3 codepoints. The large drop from all words to $\geq$3cp for P and S is partly the short-word `img_to_ink` artefact.

Source: `runs/R_eval/RESULTS.md`

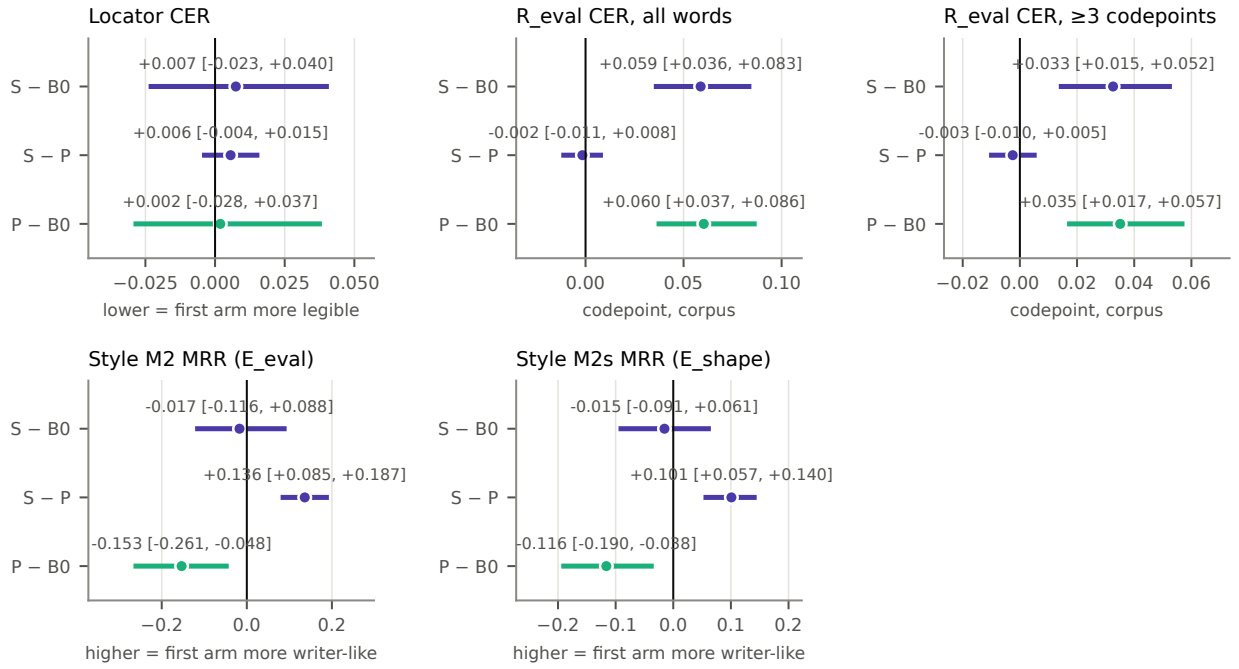


Figure 33: Paired differences S−B0, S−P and P−B0, with 95% writer-bootstrap CIs, for CER (top) and style (bottom). The locator cannot separate the arms. R_eval shows S and P less legible than B0 and tied with each other. Style shows S tied with B0 and clearly better than P. *The CER panels are in different units.*

Source: `runs/P1a3/eval/metrics.json`, `runs/R_eval/RESULTS.md`

How to read the legibility gap.

- **The short-word artefact explains only part of it.** S's CER is 0.089 on all words and 0.055 on ≥3cp words. The all-word gap to B0 ($+0.059$) is inflated by the legacy `img_to_ink` smudge on lone glyphs. The ≥3cp gap, $+0.033$ [+0.015, +0.052], **remains significant**. S really is less legible than B0, by about 1.3× real's CER on ≥3cp words. (One of the gathered notes called the gap "an artefact of short-word ink normalisation"; that overstates it.)
- **Left-alignment caveat.** A later CPU check (Opus review, `runs/S_ocrloss/REVIEW.md`, §A) found a framing mismatch. R_eval was trained on left-aligned crops, but S/P renders are centred by `fit_canvas`. On 400 cached PNGs per arm, the median left margin is 54 px for S, 6 px for real and 0 px for B0. Centring real crops raises R_font2real micro-CER from 0.047 to 0.096. Left-shifting S renders moves S only from 0.099 to 0.092 ($n = 800$). So layout explains only about 0.007 of S's ≈0.06 gap to B0; most of the gap is real legibility. These are that review's subsample numbers, not the RESULTS.md protocol.
- **CER is legibility, not fidelity.** B0 is read *more* easily than the writers' own crops (-0.0136 [-0.0268, -0.0021]). A clean, font-like render beats real handwriting under any recogniser. CER can therefore only say that an arm is legible enough. Arms are ranked on CER together with style and pen.

8 Bugs found and lessons

Most of the project's large swings in conclusions came from measurement, not from models. Table 22 collects every pitfall that changed a number or a decision.

Table 22: Measurement pitfalls and bugs, with their effect and fix.

pitfall	what happened	effect / fix
Uncapped CER (P1a)	1-character punctuation targets read as 23–29 characters; 15% of words had CER > 1	CER > 1 was an artefact; cap at 1 and exclude punctuation

pitfall	what happened	effect / fix
Grey-box failures (P1a)	writer 117's grey refs led to grey rectangles; `img_to_ink` turned them into "ink", so w/core $\approx$100	mean w/core $\approx$4 was one writer; whiten refs, then bgnorm2
Eval asymmetry (P1a)	real/B0 tight and left-aligned, generated centred on a 256 canvas; B0 $>$256 px truncated	0.04–0.05 CER bias; left-align, report it
Same noise for every word (P1a)	all words sampled from seed 0	correlated outputs; per-word seeds
CFG-null bug (P1a$'$)	content map and tokens in both CFG branches, so content was never guided	"B$'$ NO-GO, 0.550" inverted to "best arm, 0.188"; joint null
Integer EDT pen (eval_v2)	$2\,\mathrm{EDT} - 1$ takes values $\{1, 1.83, 3, \dots\}$ on 64 px	1.8–2.2$\times$ reported vs 1.13–1.36$\times$ true; sub-px/core metric
Underpowered OOV gate (P1a$'$)	15 OOV words, one word carried the effect; sign-inverted gate code	replaced by OOV on all 20 writers, $\geq$2 seeds
Off-by-one vocab size (eval_v2)	`vocab_size=len(vocab)` where training used $+1$ for pad/OOV	fixed at both call sites
bg-fail misses faint blotches	writer 117 blotches are faint (ink 0.18–0.5)	faint-blotch score added
`img_to_ink` short-word smudge	5th-percentile floor stretches paper texture to black on lone glyphs (2–3% ink)	short-word CER/blotch for P/B$'$/S inflated; `min_span=100` (opt-in), caches not yet rescored
Gradient checkpoint bug (S)	side-channel attributes reset before the checkpoint recompute	S diverged twice; fixed and tested
Cached PNGs are mattes	saved as $255(1 - \text{ink})$; re-running `crop_ink` restretches ink	read with $1 - \text{png}/255$ (blocker fixed before R_eval)
Wrong word pairing (R_eval)	seeds differ across arms (B0/real 1, diffusion 2)	merge on (dataset, writer, idx, seed_k), never zip
Augmentation cut words (R_eval)	symmetric $\pm$12% roll deleted word edges	roll only into empty columns
B0 wider than 256 px	$\approx$10% of B0 words, up to 628 px	locator truncates, R_eval squashes: B0 rows differ between the two
Locator vs R_eval units	letter-unit capped per-word mean vs codepoint corpus ratio	never compare across the two
Recogniser CER $\neq$ fidelity	B0 beats real crops under R_eval	rank arms with style and pen too
R_eval framing	R trained left-aligned, S/P centred	$\approx$0.007 of S's gap; left-shift in future evals
Biased e2e C2ST (layout)	pooled OOF AUC with one label per group; floor 0.34 / 0.115	e2e_unfit, e2e_unfit_ink C2ST not comparable; balanced folds
Shuffled real as "floor"	breaks gap–width relation	real null is within-page split, 0.496
Per-word wobble overshoot	line-scale inflation applied to every word	up to 2$\times$ wobble; leverage correction
Selection on TEST (layout)	every choice compared on TEST	optimistic by $\sim$1 sd; VAL check added
Planner history across pages	history 56–72 lines vs $\leq$16 in training; slope random walk	per-page reset
Survivor bias in e2e	detector splits steep/bowed lines; metrics see survivors	slope/drift W1 not comparable across runs
Ink-extent threshold	`INK_LEVEL` 0.35 vs page Otsu $\approx$0.39	$+0.067$ U gap bias
Uncontrolled 2-D FFT	block height alone separates real from generated (AUC 0.985)	only the controlled test is valid
Fourier "17$\times$"	compared an unrealisable continuous trace with a floor artefact	fair: Fourier 0.668 vs v1 0.702

pitfall	what happened	effect / fix
Vacuous tests (layout)	noise-floor tolerance $\lvert\text{AUC}-0.5\rvert < 0.4$; tests on fake models	tightened to 0.15; 70 tests

Source: `runs/P1a/REVIEW.md`, `runs/P1a2/REVIEW.md`, `EXPERIMENTS.md` (Known pitfalls), `runs/S_ocrloss/REVIEW.md` §A (framing only), `scribe_layout_v2/report/FACTS.md`, `scribe_layout_v2/runs/review_opus.md`

Lessons.

1. **Review the sampler and the metric before the model.** Two verdicts (P1a′ B′, the pen ratio) flipped because of the evaluation code, not the model.
2. **Every metric needs a null and a ceiling measured in the same pipeline**: real-vs-real C2ST, real-crop CER, real MRR, fonts as the style null. A null of 0.34 instead of 0.5 is a protocol failure, not a result.
3. **Measure like with like.** Sub-pixel width relative to letter size, the same canvas convention, the same framing as the recogniser's training data.
4. **Keep a selection-free split.** TEST-selected layout gains shrank or vanished on VAL.
5. **Legibility is necessary but not sufficient.** A recogniser rewards font-like renders, so it cannot rank fidelity.

8.1 Contradictions between sources

- **Fourier "17×".** `scribe_layout_fourier/runs/results.md` still states it. `FACTS.md` corrects it with the fair comparison (0.668 vs 0.702). This report follows FACTS.
- **fft2d uncontrolled section.** The results file reports high AUCs that are confounded; only its controlled section is used here.
- **R_eval row label (resolved).** `runs/R_eval/RESULTS.md` lists `Bp_t5c5` (0.1395 / 0.1021) but not B′v2; the full table in `runs/R_eval/full_chain.log` scores both, and `Bv2_t5c5` is 0.1412 / 0.1024, as in `EXPERIMENTS.md`.
- **The 41 BW words missing from the R_eval cache (resolved).** An early attribution to a Unicode nukta mismatch was wrong: they are the 41 `measure_rows` failures of the T0 index (`runs/T0/REPORT.md` §4.2).
- **Writer disjointness in hcr/996.** The v1 layout report says TEST lines are by unseen writers. `FACTS.md` says writers likely repeat across texts and that this cannot be verified (no writer IDs).
- **The gathered notes** contain errors that this report does not repeat. Note 04 calls S's R_eval gap an artefact (only part of it is). It says the second diverged S run was stable (its rolling loss rose to 0.108 by step 1,050). It says "CER capped" for R_eval (the corpus CER is not capped). Note 05 gives a TEST paired diff that is not in any source (dropped). Note 03 mislabels P1a_B as A′. Note 06 has invented column lists.

9 Discussion

What works.

- **Learned layout.** The planner is small (seven 1-D MDNs) and trains in seconds. It reproduces real spacing dependence (gap lag-1, gap–width, slope persistence) and brings line C2ST from 0.98 to about 0.53 at plan level. It is integrated in BS with clearance, margins and per-page resets.
- **Content through tokens.** Grapheme tokens in cross-attention carry word identity (content-only CER 0.199); a painted map does not.
- **Reference tokens for style.** S closes the style gap to B0 (-0.017, n.s.) and beats every other diffusion arm by about 0.14 MRR.
- **Pen.** Rebalancing plus a pen input brings the raw ratio from 1.29–1.30 (B′) to 1.13–1.15. Post-hoc calibration reaches 1.06.
- **Fonts as a recogniser prior**, especially with little real data (0.178→0.115 with 10% of writers).

What does not (yet).

- **Legibility of the diffusion arms.** On words of $\geq$3 codepoints S and P are about 1.3$\times$ real's R_eval CER and 2.4$\times$ B0's. This is now the main gap.
- **S does not beat B0 on style.** It ties. On BN-HTR, B0 is close to real (0.883 vs 0.916); on BW all arms are far from real (S 0.523, B0 0.500, real 0.789).
- **The pen knob in S is inert** (slope 0.03), so pen control must come from calibration.
- **Page-level integration.** Text renders small on BW pages; P draws dashes as bars; short words carried the smudge artefact.
- **Layout gaps.** x_{start} is the weakest learned quantity. End-to-end C2ST (0.72 / 0.62 with context) is far from the 0.53 floor, partly because of extraction survivor effects.

Open problems. (1) Legibility without losing style. A recogniser-guided objective is the obvious lever, but it must not reuse R_eval, which would make the evaluation circular. (2) Thin pens like image_2 (1.6 px at 64 px) need a larger canvas; the review recommended 128×512 `fit_canvas`. (3) The latent writer-style layout model's context mismatch. (4) A correlated-wobble calibration for the per-word unfit correction. (5) The hcr writer-identity question, which limits both layout splits and any hcr content-matched style metric.

10 Next steps

1. **Running now (no results cited here):**
   - *smudge-fix rescore*: rescore all arms together with `min_span` and re-extracted inks, removing the short-word artefact from the S/P vs B0 gap;
   - *best-of-N recogniser selection*: sample several candidates per word and keep the most legible;
   - *recogniser-loss fine-tuning*: fine-tune S with an auxiliary CTC loss. Its review asks that it be scored both as-is and left-aligned, with a left-margin diagnostic, so that a "move ink left" shortcut is not mistaken for legibility.
2. Add left-aligned R_eval scoring as a standard column for every generator arm.
3. Fix BW page scale and dash words at integration; re-render the page sheets.
4. Revive pen control in S (knob dropout or a stronger embedding), or adopt calibration as the official pen path.
5. P1b at 128×512 `fit_canvas`, warm-started from S, for thin-pen writers.
6. Per-writer test-time adaptation (LoRA) with leave-crop-out stopping and the M6 copy check.
7. **Checkpoint C**: one TEST evaluation of the final configuration against B0 and the baselines, with all metrics.

A Reproduction

Python is `/home/hpc4090/shyan/ocr/venv/bin/python`. Run one GPU job at a time under the shared lock `runs/.gpu.lock`. Working directory: `scribe_wordgen/` unless noted.

step	command / path
this report	`cd reports/final && python make_figs.py && latexmk -xelatex report.tex`
data index, splits	`wordgen/data/index.py` → `data_cache/index_{bnhtr,bw}.csv`, `data_cache/splits.json`
BW crops (missing on disk)	`wordgen/recog/bw_regen.py` (in memory, from `banglawriting/converted/`)
eval_v3	`python -m wordgen.eval.eval_v3 --stage gen|import|score|validate|report|sheets` → `runs/eval_v3/`
style metric	`python -m wordgen.eval.style_metric --stage all` → `runs/style_eval/`
P/S training	`wordgen/train/train_diffusion3.py`; checkpoints `runs/P1a3/{P,S}/unet3_final.pt`
P/S evaluation	`wordgen/eval/eval_p1a3.py` → `runs/P1a3/eval/report_tables.txt`
pen calibration	`wordgen/pen/calibrate.py` → `runs/pen_calib/`
R_eval chain	`runs/R_eval/full_chain.sh` (train, select, TEST once, rescore) → `runs/R_eval/RESULTS.md`
page sheets	`wordgen/integrate/page_sheets.py` → `runs/page_sheets/sheets/`
layout model	`cd scribe_layout_v2`; train with `configs/default_unfitw_dc.yaml`; default `runs/default_unfitw_dc/model.pt`
layout 5-seed compare	`scribe_layout_v2/runs/compare_unfitw/compare_5seed*.py`
layout end to end	`scribe_layout_v2/runs/e2e_final/` (results.md, results.json)

B Glossary

B0 BengaliScribe's shipped font-prior GAN word generator.

B3 DiffusionPen fine-tuned on all BW writers (`dp_words_b64`); a probe/baseline only.

A′, B′ P1a′ diffusion arms: text only, and text + grapheme tokens.

P, S P1a3 arms: B′ + pen knob + null style (P); + reference tokens in decoupled cross-attention (S).

tXcY text guidance X, content guidance Y in classifier-free guidance.

CFG null the unconditional branch; must drop text, map and tokens jointly.

core, U body height under the matra; U = a page's median word core height (layout unit, ≈3.5 mm).

matra the Bengali headline stroke.

locator CER CER from BS's `attn_locator`; letter units, capped per-word mean.

R_eval the independent CRNN-CTC recogniser (R_font2real); codepoint corpus CER.

cp codepoint.

M2 / M2s writer-ID MRR with E_eval (shape + pen) / E_shape (skeleton only).

MRR mean reciprocal rank of the true writer among candidates.

pen ratio median over writers of (generated / own real) sub-pixel width per core.

bg-fail, blotch background-failure and faint-blotch rates.

OOV target word not in the training texts (for R_eval: also split by the synthetic vocabulary).

C2ST classifier two-sample test; AUC 0.5 = indistinguishable.

W1, KS Wasserstein-1 distance and Kolmogorov–Smirnov statistic between marginals.

MDN mixture-density network head.

unfit / unfitw / dc layout targets inflated for fit shrinkage (line table / per-word leverage) / curvature deconvolution.

e2e end to end: BS render → real detector → same extraction.

VAL / TEST writer-disjoint (words) or text-disjoint (layout) evaluation splits.